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MIL-M-28787/514
15 December 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY MAPPED I/O BUS INTERFACE MODULE, KEY CODE CWA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Memory Mapped Input/Output (MMIO) bus interface module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5963

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.535 failures per million operating hours for class I or 3.601 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be CWA.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Voc stress	IV	S	---------Module shall pass all remaining tests---------						
D1-D4	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D5-D8	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D9	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D10-D13	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D14,D15	Ibd	V	S	-0.58	0.68	-0.58	0.68	-0.58	0.68	mA
D16-D21	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D22	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D23	Ibd	V	S	-0.58	1.67	-0.58	1.67	-0.58	1.67	mA
D24-D29	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D30-D32	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D33-D35	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D36	Ibd	V	S	0.0	4.0	0.0	4.0	0.0	4.0	mA
D37	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D38	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D39	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D40	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
E1-E65	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E66-E70	Ios	VI	S	-225.0	-100.0	-225.0	-100.0	-225.0	-100.0	mA
E71,E72	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	mA
E73	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1-F4	Iih	VII	X	-1.56	0.06	-1.56	0.06	-1.56	0.06	mA
F5-F8	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F9-F16	Iih	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F17	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F18-F21	Iih	VII	X	-0.96	0.04	-0.96	0.04	-0.96	0.04	mA
F22,F23	Iih	VII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F24-F29	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F30	Iih	VII	X	-1.32	0.06	-1.32	0.06	-1.32	0.06	mA
F31	Iih	VII	X	-28.1	-14.3	-28.1	-14.3	-28.1	-14.3	mA
F32	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F33-F37	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F38-F40	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F41	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F42-F49	Iih	VII	X	-1.3	0.14	-1.3	0.14	-1.3	0.14	mA
F50	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F51	Iih	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F52	Iih	VII	X	-1.6	0.02	-1.6	0.02	-1.6	0.02	mA
F53	Iih	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F54	Iih	VII	X	-1.6	0.2	-1.6	0.2	-1.6	0.2	mA
F55	Iih	VII	X	-1.6	0.02	-1.6	0.02	-1.6	0.02	mA
F56	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F57	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F58	Iih	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G1-G4	Iil	VIII	X	-1.56	0.06	-1.56	0.06	-1.56	0.06	mA
G5-G8	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G9-G16	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G17	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G18-G21	Iil	VIII	X	-0.96	0.04	-0.96	0.04	-0.96	0.04	mA
G22,G23	Iil	VIII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA
G24-G29	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G30	Iil	VIII	X	-1.32	0.06	-1.32	0.06	-1.32	0.06	mA
G31	Iil	VIII	X	-28.1	-14.3	-28.1	-14.3	-28.1	-14.3	mA
G32	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G33-G37	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G38-G40	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G41	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G42-G49	Iil	VIII	X	-1.3	0.14	-1.3	0.14	-1.3	0.14	mA
G50	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G51	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G52	Iil	VIII	X	-1.6	0.02	-1.6	0.02	-1.6	0.02	mA
G53	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G54	Iil	VIII	X	-1.6	0.2	-1.6	0.2	-1.6	0.2	mA
G55	Iil	VIII	X	-1.6	0.02	-1.6	0.02	-1.6	0.02	mA
G56	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G57	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G58	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
H1-H73	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J66	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K8	Iozh	XI	X	-0.1	0.1	-0.1	0.1	-0.1	0.1	mA
K9	Iozh	XI	X	0.0	0.25	0.0	0.25	0.0	0.25	mA
K10-K17	Iozh	XI	X	-0.1	0.1	-0.1	0.1	-0.1	0.1	mA
K18	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K19-K34	Iozl	XI	X	-0.1	0.1	-0.1	0.1	-0.1	0.1	mA
K35	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1-L20	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L21	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L22-L30	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L31	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L32-L37	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	7.1	13.2	7.1	13.2	7.1	13.2	uF
M2	PSFC	XIII	S	7.1	13.3	7.1	13.3	7.1	13.3	uF
N	Icc	XIV	X	500	1200	500	1200	500	1200	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	Class I and II 25 ± 5°C 4/		Class I 0 ± 5°C 4/ 60 ± 5°C		Class II -55 ± 5°C 4/ 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R1	Ci/o	XVIII	Q	6	125	6	125	6	125	pF
R2,R3	Ci/o	XVIII	Q	6	100	6	100	6	100	pF
R4	Ci/o	XVIII	Q	6	125	6	125	6	125	pF
R5-R7	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R8	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R9	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R10-R12	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R13-R15	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R16	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R17	Ci/o	XVIII	Q	2	50	2	50	2	50	pF
R18-R21	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R22	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R23	Ci/o	XVIII	Q	6	125	6	125	6	125	pF
R24	Ci/o	XVIII	Q	2	100	2	100	2	100	pF
R25-R29	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R30-R32	Ci/o	XVIII	Q	2	125	2	125	2	125	pF
R33-R35	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R36	Ci/o	XVIII	Q	8	100	8	100	8	100	pF
R37	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R38	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R39,R40	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R41-R48	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
R49	Ci/o	XVIII	Q	2	150	2	150	2	150	pF
R50,R51	Ci/o	XVIII	Q	2	125	2	125	2	125	pF
R52	Ci/o	XVIII	Q	2	150	2	150	2	150	pF
R53	Ci/o	XVIII	Q	2	125	2	125	2	125	pF
R54	Ci/o	XVIII	Q	2	150	2	150	2	150	pF
R55,R56	Ci/o	XVIII	Q	2	125	2	125	2	125	pF
R57	Ci/o	XVIII	Q	4	125	4	125	4	125	pF
R58	Ci/o	XVIII	Q	4	75	4	75	4	75	pF
T1,T2	PSTA	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/514-1 for class I and M28787/514-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (V_{CC}) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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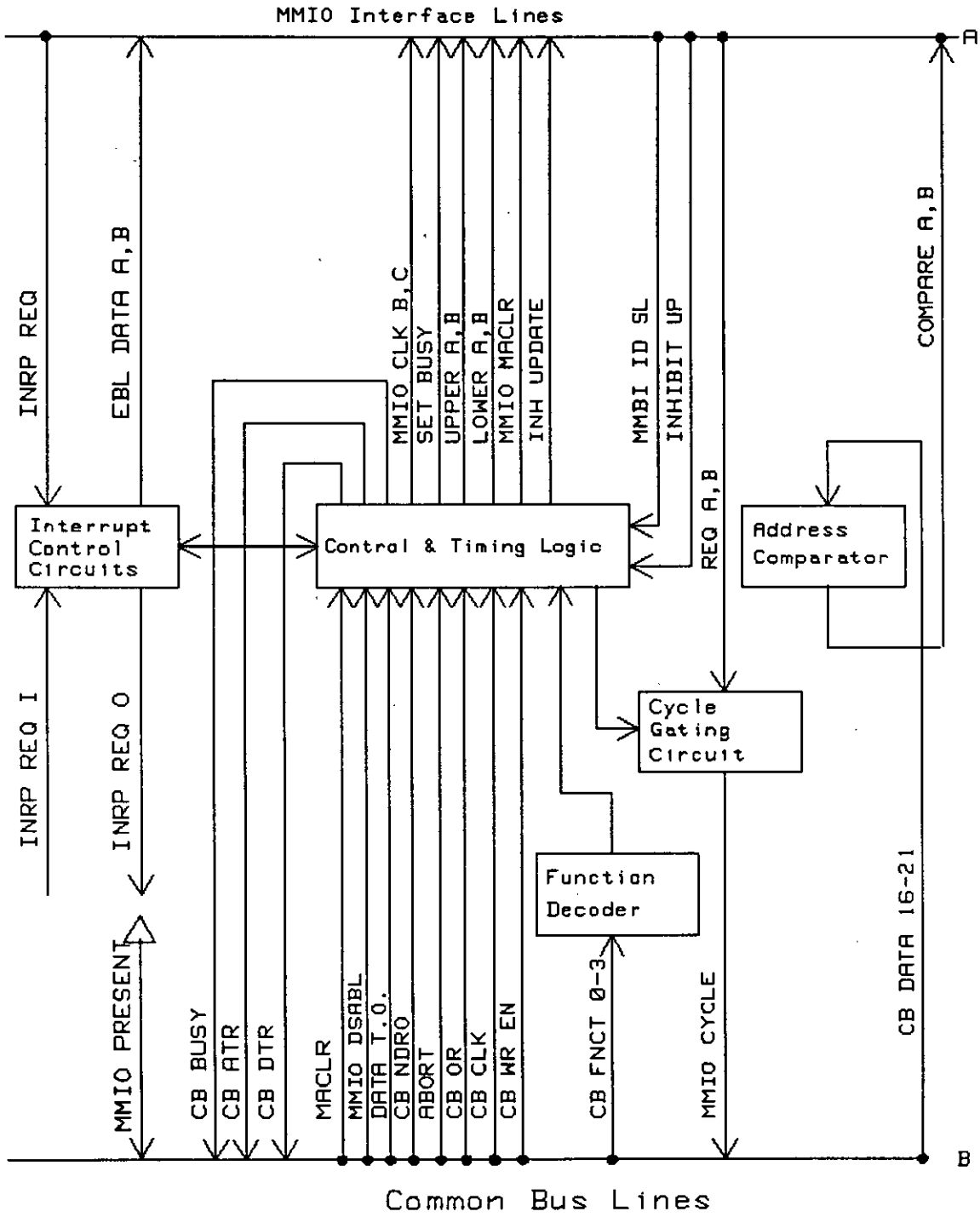


FIGURE 1. CWA functional block diagram.

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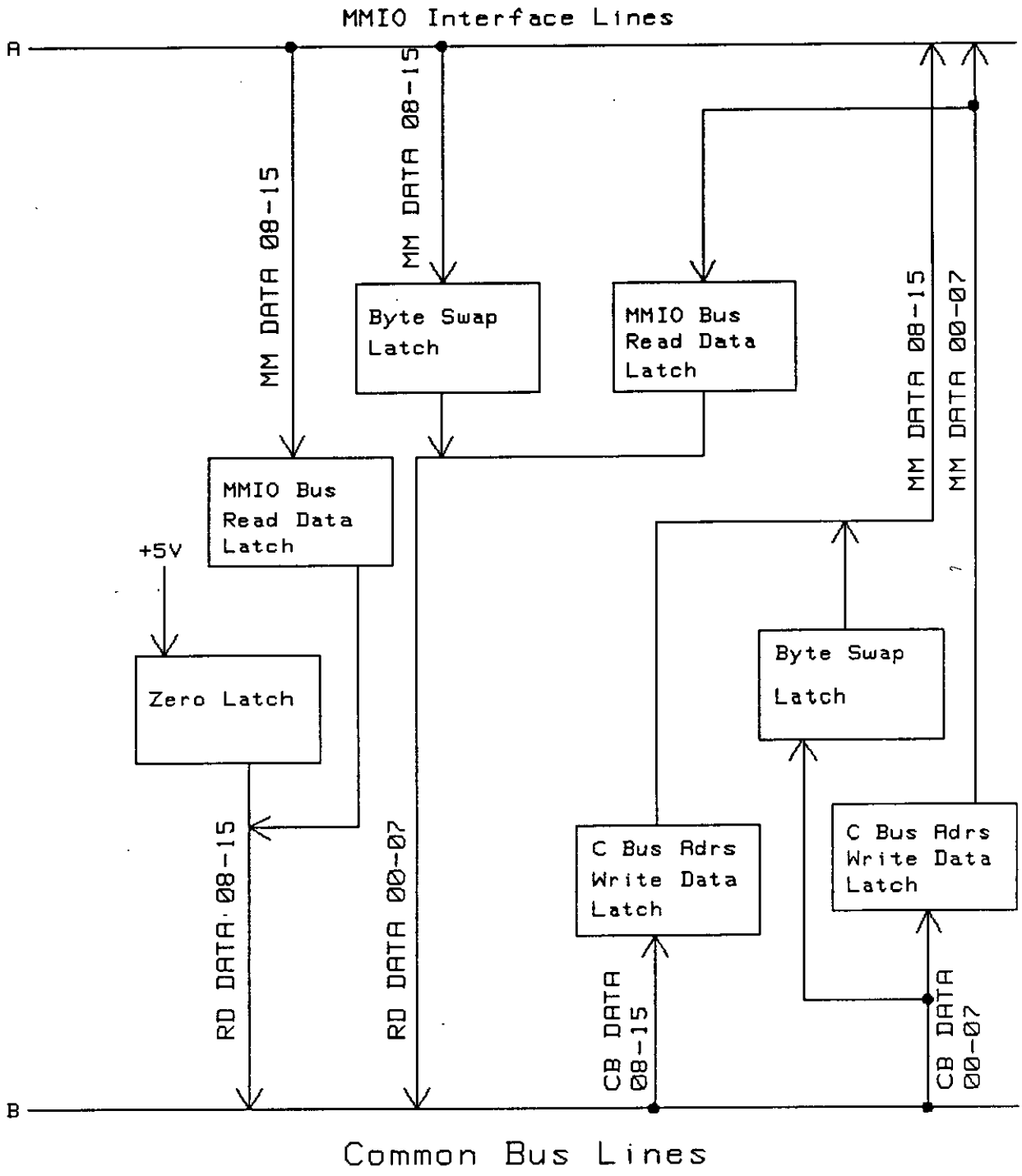


FIGURE 1. CWA functional block diagram - Continued.

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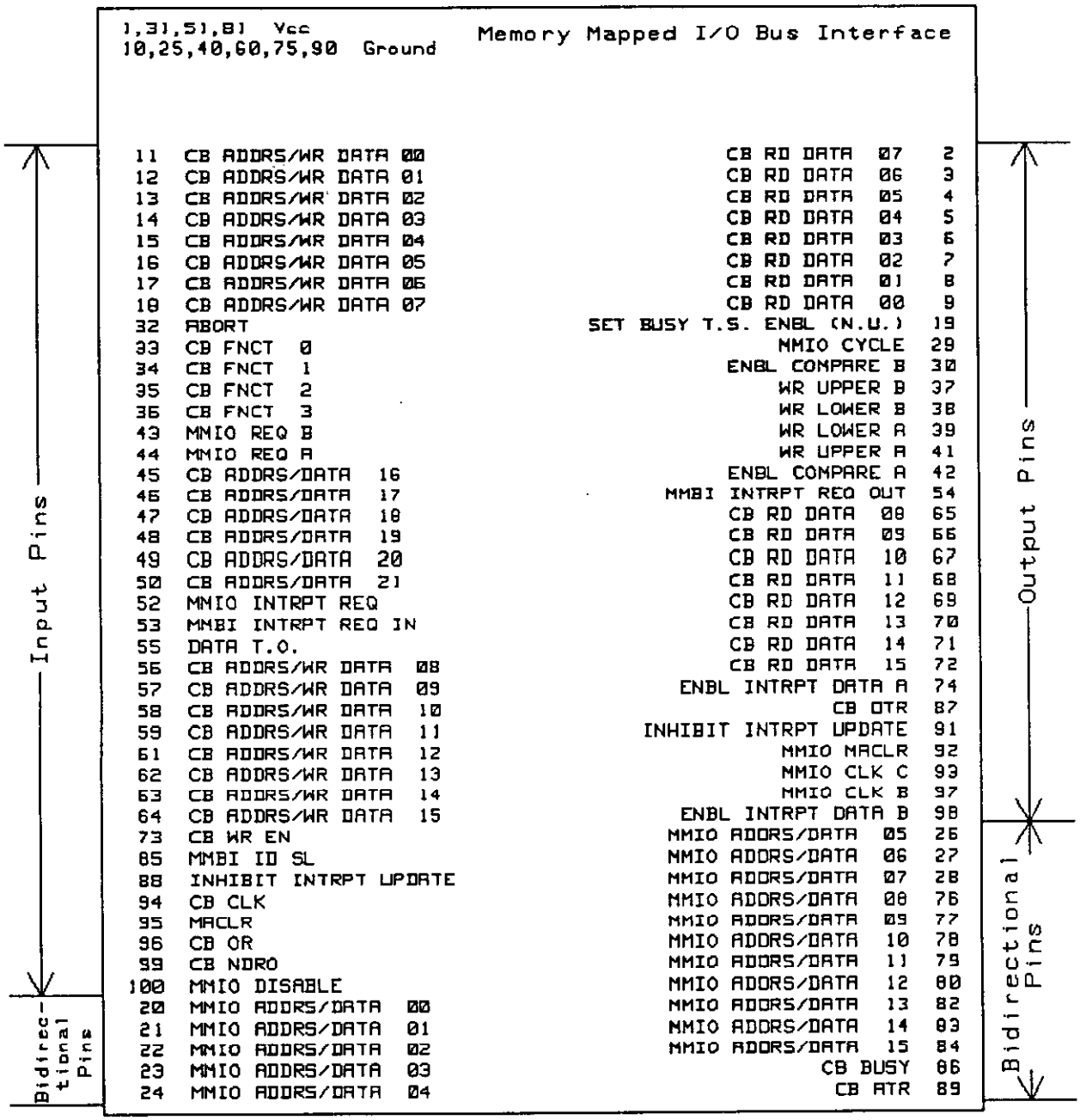


FIGURE 2. CWA pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

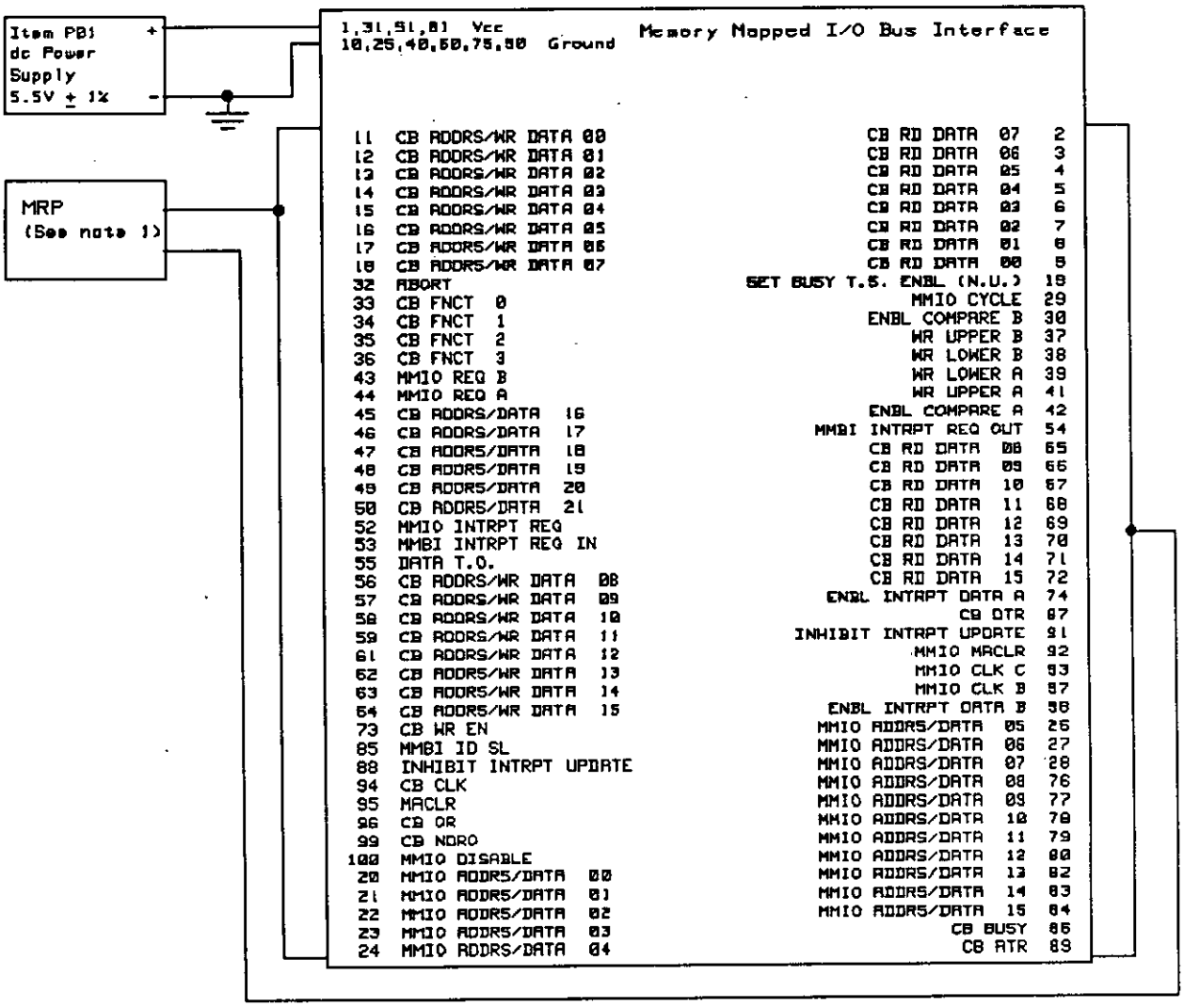
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. CWA operating temperature and component temperature test circuit.

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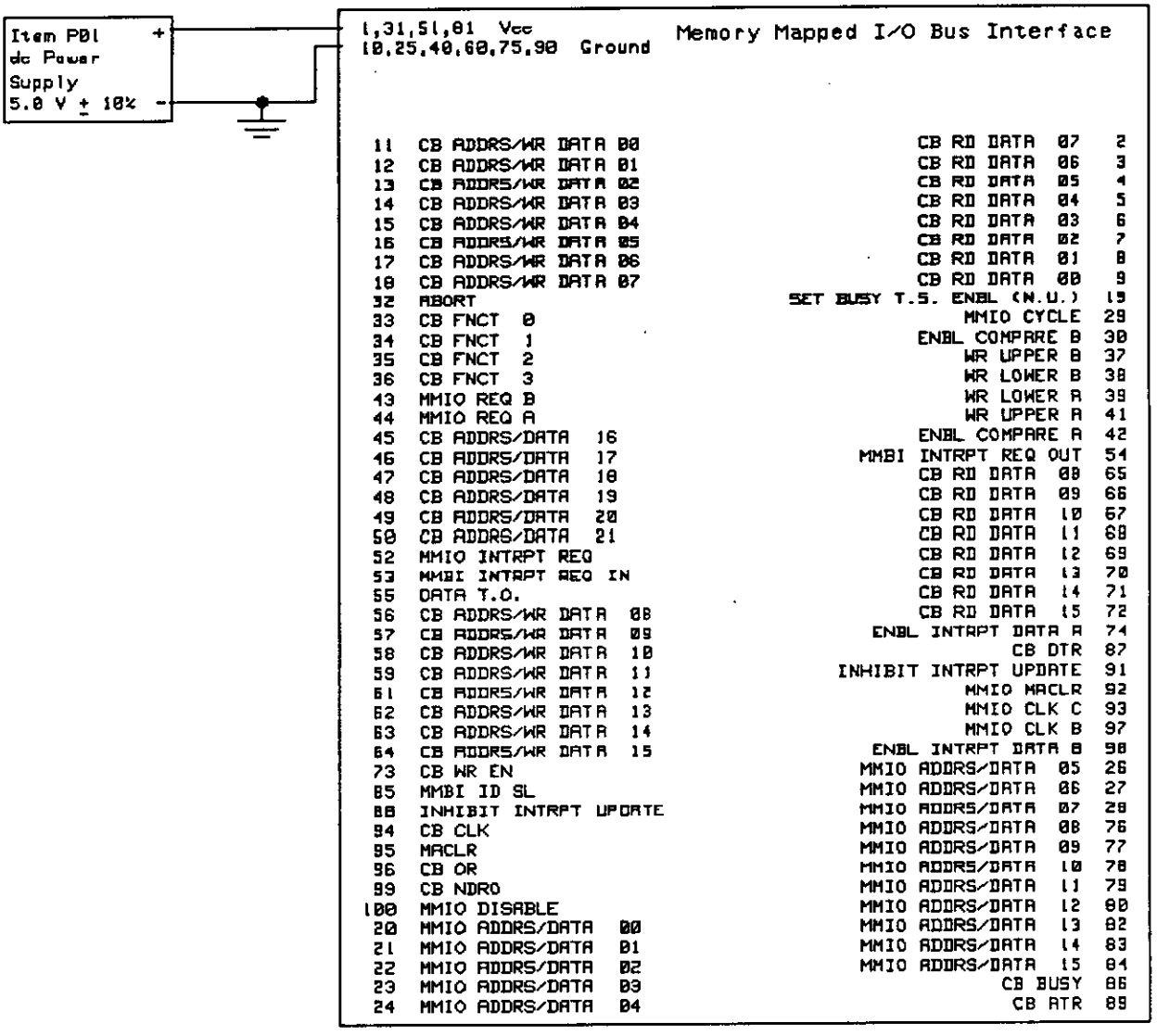


FIGURE 4. CWA vibration test circuit.

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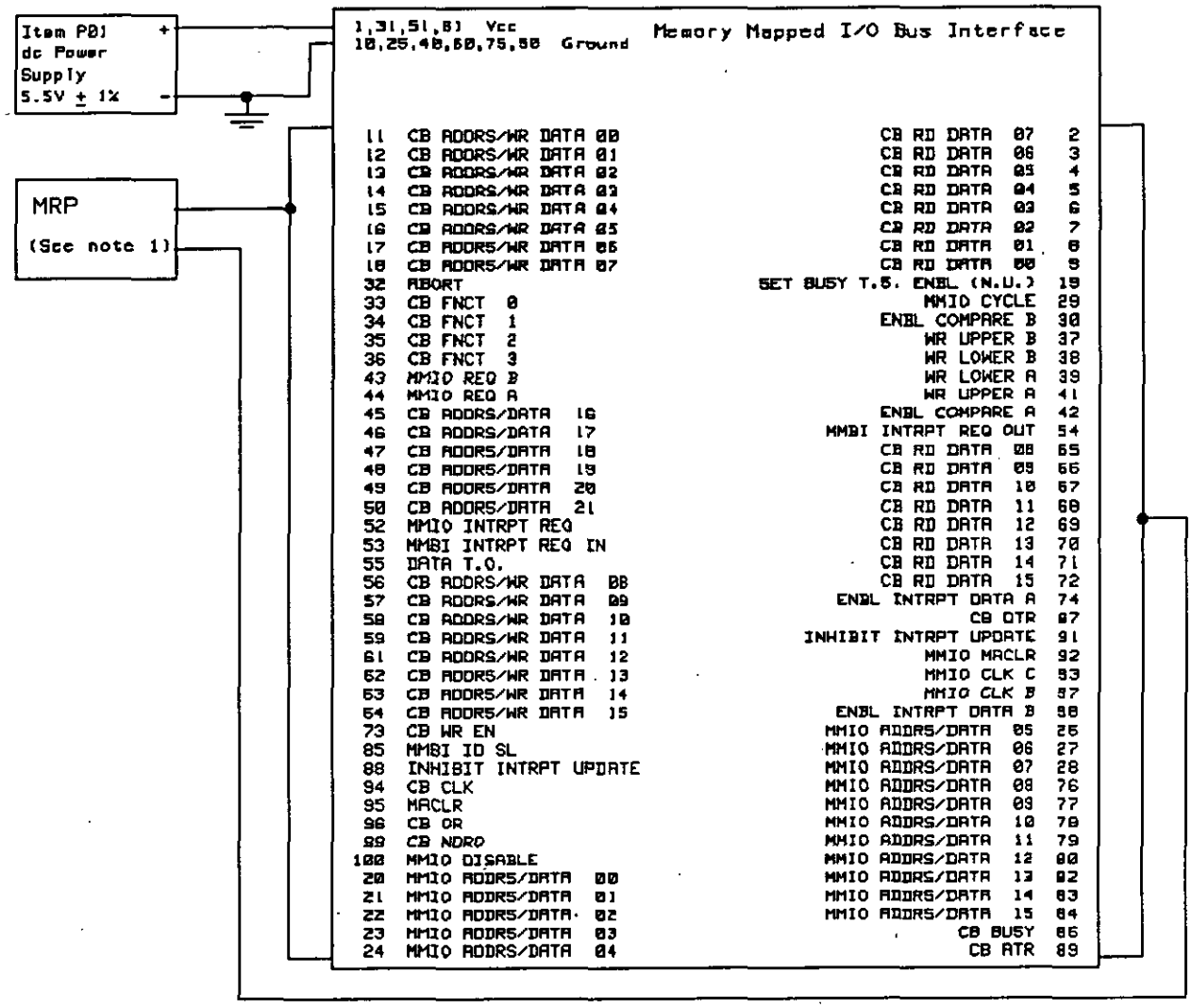
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. CWA life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 6.60 W (max); 4.43 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB DATA 00	-1.2	0.4	2.4	-1.56	0.06	0.3	125
CB DATA 01	-1.2	0.4	2.4	-1.56	0.06	0.3	100
CB DATA 02	-1.2	0.4	2.4	-1.56	0.06	0.3	100
CB DATA 03	-1.2	0.4	2.4	-1.56	0.06	0.3	125
CB DATA 04	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB DATA 05	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB DATA 06	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB DATA 07	-1.2	0.4	2.4	-1.2	0.04	0.2	100
ABORT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB FNCT 0	-1.2	0.4	2.4	-0.96	0.04	0.2	75
CB FNCT 1	-1.2	0.4	2.4	-0.96	0.04	0.2	75
CB FNCT 2	-1.2	0.4	2.4	-0.96	0.04	0.2	75
CB FNCT 3	-1.2	0.4	2.4	-0.96	0.04	0.2	100
REQ B	NA	0.4	2.4	-27.7	-14.3	0.68	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
REQ A	NA	0.4	2.4	-27.7	-14.3	0.68	100
CB DATA 16	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB DATA 17	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB DATA 18	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB DATA 19	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB DATA 20	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB DATA 21	-1.2	0.4	2.4	-0.6	0.02	0.1	75
INRP REQ	-1.2	0.4	2.4	-1.32	0.06	0.2	100
INRP REQ I	NA	0.4	2.4	-28.1	-14.3	1.67	125
DATA T.O.	-1.5	0.4	2.4	-0.2	0.02	0.1	100
CB DATA 08	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB DATA 09	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB DATA 10	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB DATA 11	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB DATA 12	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB DATA 13	-1.2	0.4	2.4	-1.2	0.04	0.2	125
CB DATA 14	-1.2	0.4	2.4	-1.2	0.04	0.2	125
CB DATA 15	-1.2	0.4	2.4	-1.2	0.04	0.2	125
CB WR EN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MMBI ID SL	-1.5	0.4	2.4	-0.2	0.02	0.1	75
INHIBIT UP	-1.2	0.4	2.4	-1.6	0.02	0.1	75
CB CLK	-1.2	0.4	2.4	-1.6	0.2	4.0	100
MACLR	-1.2	0.4	2.4	-1.6	0.02	0.1	75
CB OR	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB NDRO	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MMIO DSABL	-1.2	0.4	2.4	-0.8	0.04	0.2	75
MM DATA 00	NA	0.4	2.4	-0.65	0.07	NA	75
MM DATA 01	NA	0.4	2.4	-0.65	0.07	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MM DATA 02	NA	0.4	2.4	-0.65	0.07	NA	75
MM DATA 03	NA	0.4	2.4	-0.65	0.07	NA	75
MM DATA 04	NA	0.4	2.4	-0.65	0.07	NA	75
MM DATA 05	NA	0.4	2.4	-0.65	0.07	NA	75
MM DATA 06	NA	0.4	2.4	-0.65	0.07	NA	75
MM DATA 07	NA	0.4	2.4	-0.65	0.07	NA	75
MM DATA 08	NA	0.4	2.4	-1.3	0.14	NA	150
MM DATA 09	NA	0.4	2.4	-1.3	0.14	NA	125
MM DATA 10	NA	0.4	2.4	-1.3	0.14	NA	125
MM DATA 11	NA	0.4	2.4	-1.3	0.14	NA	150
MM DATA 12	NA	0.4	2.4	-1.3	0.14	NA	125
MM DATA 13	NA	0.4	2.4	-1.3	0.14	NA	150
MM DATA 14	NA	0.4	2.4	-1.3	0.14	NA	125
MM DATA 15	NA	0.4	2.4	-1.3	0.14	NA	125

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB BUSY	NA	0.4	2.4	-0.65	0.07	NA	125
CB ATR	NA	0.4	2.4	-0.65	0.07	NA	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
RD DATA 07	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 06	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 05	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 04	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 03	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 02	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 01	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 00	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
SET BUSY	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
MMIO CYCLE	N/A	0.5	-0.4	6.0	0.25	NA	NA	NA
COMPARE B	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
UPPER B	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
LOWER B	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
LOWER A	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
UPPER A	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
COMPARE A	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
INRP REQ O	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
RD DATA 08	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 09	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 10	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 11	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 12	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 13	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 14	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
RD DATA 15	2.4	0.5	-0.4	6.0	0.1	0.1	-60.0	NA
EBL DATA A	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CB DTR	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
INH UPDATE	2.4	0.5	-0.4	6.0	NA	NA	-100.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MMIO MACLR	2.4	0.5	-0.4	6.0	NA	NA	-100.0	NA
MMIO CLK C	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
MMIO CLK B	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
EBL DATA B	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
MM DATA 00	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MM DATA 01	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MM DATA 02	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MM DATA 03	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MM DATA 04	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MM DATA 05	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MM DATA 06	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MM DATA 07	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
MM DATA 08	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
MM DATA 09	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
MM DATA 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
MM DATA 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
MM DATA 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
MM DATA 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	150
MM DATA 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
MM DATA 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
CB BUSY	2.4	0.5	-0.4	6.0	NA	NA	-100.0	125
CB ATR	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CB CLK to COMPARE B	-	-	-	36	36	36	nS	11
1	tpd	CB CLK to COMPARE A	-	-	-	36	36	36	nS	11
2	tpd	REQ B to MMIO CYCLE	-	-	-	27	27	27	nS	11
3	tpd	CB CLK to MMIO CYCLE	-	-	-	64	64	64	nS	11
4	tpd	CB CLK to SET BUSY	-	-	-	36	36	36	nS	11
5	tpd	CB CLK to CB BUSY	-	-	-	38	38	38	nS	11
6	tpd	CB CLK to CB DTR	-	-	-	38	38	38	nS	11
7	tpd	CB CLK to RD DATA 00	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 01	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 02	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 03	-	-	-	59	59	59	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tpd	CB CLK to RD DATA 04	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 05	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 06	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 07	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 08	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 09	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 10	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 11	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 12	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 13	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 14	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 15	-	-	-	59	59	59	nS	11
8	tpd	CB FNCT 2 to LOWER A	-	-	-	27	27	27	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	CB FNCT 2 to LOWER B	-	-	-	27	27	27	nS	11
8	tpd	CB FNCT 2 to UPPER A	-	-	-	27	27	27	nS	11
8	tpd	CB FNCT 2 to UPPER B	-	-	-	27	27	27	nS	11
9	tpd	CB CLK to LOWER A	-	-	-	47	47	47	nS	11
9	tpd	CB CLK to LOWER B	-	-	-	47	47	47	nS	11
9	tpd	CB CLK to UPPER A	-	-	-	47	47	47	nS	11
9	tpd	CB CLK to UPPER B	-	-	-	47	47	47	nS	11
10	tpd	CB DATA 00 to MM DATA 00	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 01 to MM DATA 01	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 02 to MM DATA 02	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 03 to MM DATA 03	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 04 to MM DATA 04	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 05 to MM DATA 05	-	-	-	16	16	16	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tpd	CB DATA 06 to MM DATA 06	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 07 to MM DATA 07	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 08 to MM DATA 08	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 09 to MM DATA 09	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 10 to MM DATA 10	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 11 to MM DATA 11	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 12 to MM DATA 12	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 13 to MM DATA 13	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 14 to MM DATA 14	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 15 to MM DATA 15	-	-	-	16	16	16	nS	11
11	tpd	CB CLK to MM DATA 00	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 01	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 02	-	-	-	65	65	65	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	CB CLK to MM DATA 03	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 04	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 05	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 06	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 07	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 08	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 09	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 10	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 11	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 12	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 13	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 14	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 15	-	-	-	65	65	65	nS	11

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6.4 Module function. Memory Mapped Input/Output (MMIO) channels provide data transfers between the MRP and the external equipment (without using Input/Output Channel Adapters (IOCAs)), allowing the Central Processor (CP) to use the same instruction for Input/Output (I/O) transfers as it does for memory transfers. (The I/O device appears as a memory location to the CP or Input/Output Controller (IOC)). Up to 64 MMIO channels may be implemented in the MRP. Each MMIO channel operates independently and is contained on one module, hereafter referred to as MMIO. An MMIO bus interface module (CWA) provides the interface between 1 to 16 MMIOs and the MRP common bus. An MMIO contains four registers: output data, input data, external interrupt data, and control/status data. Data and status information are exchanged between the external equipment and the MRP (via a Transistor Transistor Logic (TTL) compatible interface) under control of the external equipment (at the rate and in the mode defined by the external equipment; either 16-bit parallel or serial). Each MMIO register has an assigned address within the first 8K of main memory (0000-1FFF) which the external equipment is required to access. Main memory addresses within this range are not used and will be inhibited when MMIO is referenced. Based upon the condition of the control/status register, the MMIO channel generates an interrupt to indicate the completion of a data transfer or the existence of an external equipment discrete interrupt condition. The interrupt code and the MMIO channel identification are presented to the CP over the common bus.

6.4.1 Memory mapped input/output bus interface. The Memory Mapped Bus Interface (MMBI) module interfaces 16 MMIO channels to the common bus by detecting an address within the 0 to 8K range (generating a memory interface inhibit), controlling MMIO class III interrupts, and performing common bus handshaking and protocol. The module contains interrupt control circuits, address and data latches, an address compare detector, cycle gating circuits, a function decoder, control and timing circuits, and byte swap latches.

6.4.1.1 Address compare detector. The address compare detector is used to detect the 0 to 8K MMIO address range and trigger the cycle gating circuits. Common bus address bits 21-13 are applied to NAND functions and, should a memory reference occur within the 0 to 8K range, the detector sends its output to the MMIO module (BWG) enabling address comparison. Enable Compare A is sent to the lower 8 MMIO modules; Enable Compare B is sent to the upper 8 MMIO modules.

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6.4.1.2 Cycle gating circuits. The cycle gating circuits generate the MMIO cycle signal when an MMIO Request A (from MMIO modules 0 through 7) or MMIO Request B (from MMIO modules 8 through 15) signal is received. This signal inhibits a memory reference when an MMIO location is referenced. This signal is sent to the memory interface module and to Data Terminator Ready (DTR) generation circuits and the MMIO active circuits.

6.4.1.3 Function decoder. The function decoder interprets common bus function bits. The common bus function bits are latched and applied to a 3-to-8 decoder. The read and write function outputs from the decoder are applied to the read control circuits and the write control logic.

6.4.1.4 Interrupt control circuits. The interrupt control circuits receive interrupt requests from the Memory Mapped Input/Output Adapter (MMIOA) group. Additionally, in a multiple MMBI configuration, interrupt requests are received from the MMBI daisy-chain. The higher numbered MMBI has priority over the lower numbered MMBI. Interrupts from MMBIs are gated together to form the MMBI interrupt request out signal to the common bus. Interrupt data from an MMIO module within an MMBI group is enabled when the CP performs a master read of the MMIO. Common bus function bits and address/write data bits are compared against a 20 (Base 16) pattern which implies a master read of MMIO interrupts. Common bus handshaking is performed between the CP and MMIO and with an address compare, an originator ready capture, and an MMBI interrupt request in, the Enintr A/B signals are generated. Enintr Data A enables interrupt data on MMIO modules 0 through 7; Enintr Data B enables interrupt data on MMIO modules 8 through 15. Additionally, the Enintr A/B signals enable the output of the Addressed Terminator Ready (ATR) flip-flop to be placed onto the common bus. The only time MMBI generates an ATR is during the master read sequence.

6.4.1.5 Control and timing. The control and timing logic generates the common bus handshaking signals, enables the data and address latches during transfers to and from the MMIO, and generates read and write control gates sent to the MMIOs via the MMIO bus. Write controls are generated from common bus function bits and are sent to the MMIOA via the MMIO bus. Latch controls are also generated from the common bus function bits. These are used to enable common bus write data and byte swap latches. ATR generation is accomplished as a result of a master read by the CP of MMIO interrupts. The ATR is generated by the ATR flip-flop being reset. The reset condition is caused by an address/function match, a MMBI interrupt request in, Originator

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Ready (OR) capture, and the address being latched. MMBI only generates an ATR for the master read sequence; ATRs for all other MMIO common bus functions are generated by the memory interface module. The MMIO clock is the common bus clock. MMIO clock A is sent via MMIO bus to MMIO 0 through 7; MMIO clock B is sent to MMIO 8 through 15. Read controls are generated from the function decoder outputs. These are used to enable the data and byte swap latches. OR capture flip-flop clocks in the common bus OR. The flip-flop will be cleared by the Clr OR Cap signal, a derivative of the ATR. DTR logic generates the DTR as a result of the originator ready signal. Outputs from the DTR flip-flop are sent to the common bus and the master read flip-flop.

6.4.1.6 Byte swap latches. The 8-bit byte swap latches allow the lower data bits to be placed into the upper bit positions when writing to an MMIO location and placing the upper bits into the lower bit positions when reading from an MMIO location. Byte swap latches are controlled by gated function bits from the control and timing logic.

6.4.1.7 MMIO bus read data latches and common bus write data latches. The MMIO bus read data latches and common bus write data latches are used to hold the address and write data for a write to an MMIO channel; the address and read data for a read of an MMIO channel. Latches are enabled by function bits from the control and timing logic.

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6.4.1.8 Description of I/O pins.

Signal name	Pin number	I/O	Description
CB DATA 00	11	I	CB ADDRS/WR DATA 00
CB DATA 01	12	I	CB ADDRS/WR DATA 01
CB DATA 02	13	I	CB ADDRS/WR DATA 02
CB DATA 03	14	I	CB ADDRS/WR DATA 03
CB DATA 04	15	I	CB ADDRS/WR DATA 04
CB DATA 05	16	I	CB ADDRS/WR DATA 05
CB DATA 06	17	I	CB ADDRS/WR DATA 06
CB DATA 07	18	I	CB ADDRS/WR DATA 07
ABORT	32	I	ABORT
CB FNCT 0	33	I	CB FNCT 0
CB FNCT 1	34	I	CB FNCT 1
CB FNCT 2	35	I	CB FNCT 2
CB FNCT 3	36	I	CB FNCT 3
REQ B	43	I	MMIO REQ B
REQ A	44	I	MMIO REQ A
CB DATA 16	45	I	CB ADDRS/DATA 16
CB DATA 17	46	I	CB ADDRS/DATA 17
CB DATA 18	47	I	CB ADDRS/DATA 18
CB DATA 19	48	I	CB ADDRS/DATA 19
CB DATA 20	49	I	CB ADDRS/DATA 20

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6.4.1.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB DATA 21	50	I	CB ADDRS/DATA 21
INRP REQ	52	I	MMIO INTRPT REQ
INRP REQ I	53	I	MMBI INTRPT REQ IN
DATA T.O.	55	I	DATA T.O.
CB DATA 08	56	I	CB ADDRS/WR DATA 08
CB DATA 09	57	I	CB ADDRS/WR DATA 09
CB DATA 10	58	I	CB ADDRS/WR DATA 10
CB DATA 11	59	I	CB ADDRS/WR DATA 11
CB DATA 12	61	I	CB ADDRS/WR DATA 12
CB DATA 13	62	I	CB ADDRS/WR DATA 13
CB DATA 14	63	I	CB ADDRS/WR DATA 14
CB DATA 15	64	I	CB ADDRS/WR DATA 15
CB WR EN	73	I	CB WR EN
MMBI ID SL	85	I	MMBI ID SL
INHIBIT UP	88	I	INHIBIT INTRPT UPDATE
CB CLK	94	I	CB CLK
MACLR	95	I	MACLR
CB OR	96	I	CB OR
CB NDRO	99	I	CB NDRO
MMIO DSABL	100	I	MMIO DISABLE

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6.4.1.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RD DATA 07	2	O	CB RD DATA 07
RD DATA 06	3	O	CB RD DATA 06
RD DATA 05	4	O	CB RD DATA 05
RD DATA 04	5	O	CB RD DATA 04
RD DATA 03	6	O	CB RD DATA 03
RD DATA 02	7	O	CB RD DATA 02
RD DATA 01	8	O	CB RD DATA 01
RD DATA 00	9	O	CB RD DATA 00
SET BUSY	19	O	SET BUSY T.S. ENBL (N.U.)
MMIO CYCLE	29	O	MMIO CYCLE
COMPARE B	30	O	ENBL COMPARE B
UPPER B	37	O	WR UPPER B
LOWER B	38	O	WR LOWER B
LOWER A	39	O	WR LOWER A
UPPER A	41	O	WR UPPER A
COMPARE A	42	O	ENBL COMPARE A
INRP REQ O	54	O	MMBI INTRPT REQ OUT
RD DATA 08	65	O	CB RD DATA 08
RD DATA 09	66	O	CB RD DATA 09
RD DATA 10	67	O	CB RD DATA 10

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6.4.1.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RD DATA 11	68	O	CB RD DATA 11
RD DATA 12	69	O	CB RD DATA 12
RD DATA 13	70	O	CB RD DATA 13
RD DATA 14	71	O	CB RD DATA 14
RD DATA 15	72	O	CB RD DATA 15
EBL DATA A	74	O	ENBL INTRPT DATA A
CB DTR	87	O	CB DTR
INH UPDATE	91	O	INHIBIT INTRPT UPDATE
MMIO MACLR	92	O	MMIO MACLR
MMIO CLK C	93	O	MMIO CLK C
MMIO CLK B	97	O	MMIO CLK B
EBL DATA B	98	O	ENBL INTRPT DATA B
MM DATA 00	20	I/O	MMIO ADDRS/DATA 00
MM DATA 01	21	I/O	MMIO ADDRS/DATA 01
MM DATA 02	22	I/O	MMIO ADDRS/DATA 02
MM DATA 03	23	I/O	MMIO ADDRS/DATA 03
MM DATA 04	24	I/O	MMIO ADDRS/DATA 04
MM DATA 05	26	I/O	MMIO ADDRS/DATA 05
MM DATA 06	27	I/O	MMIO ADDRS/DATA 06
MM DATA 07	28	I/O	MMIO ADDRS/DATA 07

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6.4.1.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MM DATA 08	76	I/O	MMIO ADDRS/DATA 08
MM DATA 09	77	I/O	MMIO ADDRS/DATA 09
MM DATA 10	78	I/O	MMIO ADDRS/DATA 10
MM DATA 11	79	I/O	MMIO ADDRS/DATA 11
MM DATA 12	80	I/O	MMIO ADDRS/DATA 12
MM DATA 13	82	I/O	MMIO ADDRS/DATA 13
MM DATA 14	83	I/O	MMIO ADDRS/DATA 14
MM DATA 15	84	I/O	MMIO ADDRS/DATA 15
CB BUSY	86	I/O	CB BUSY
CB ATR	89	I/O	CB ATR

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT, AV
Navy - AS, MC
Air Force - 13, 17, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0046)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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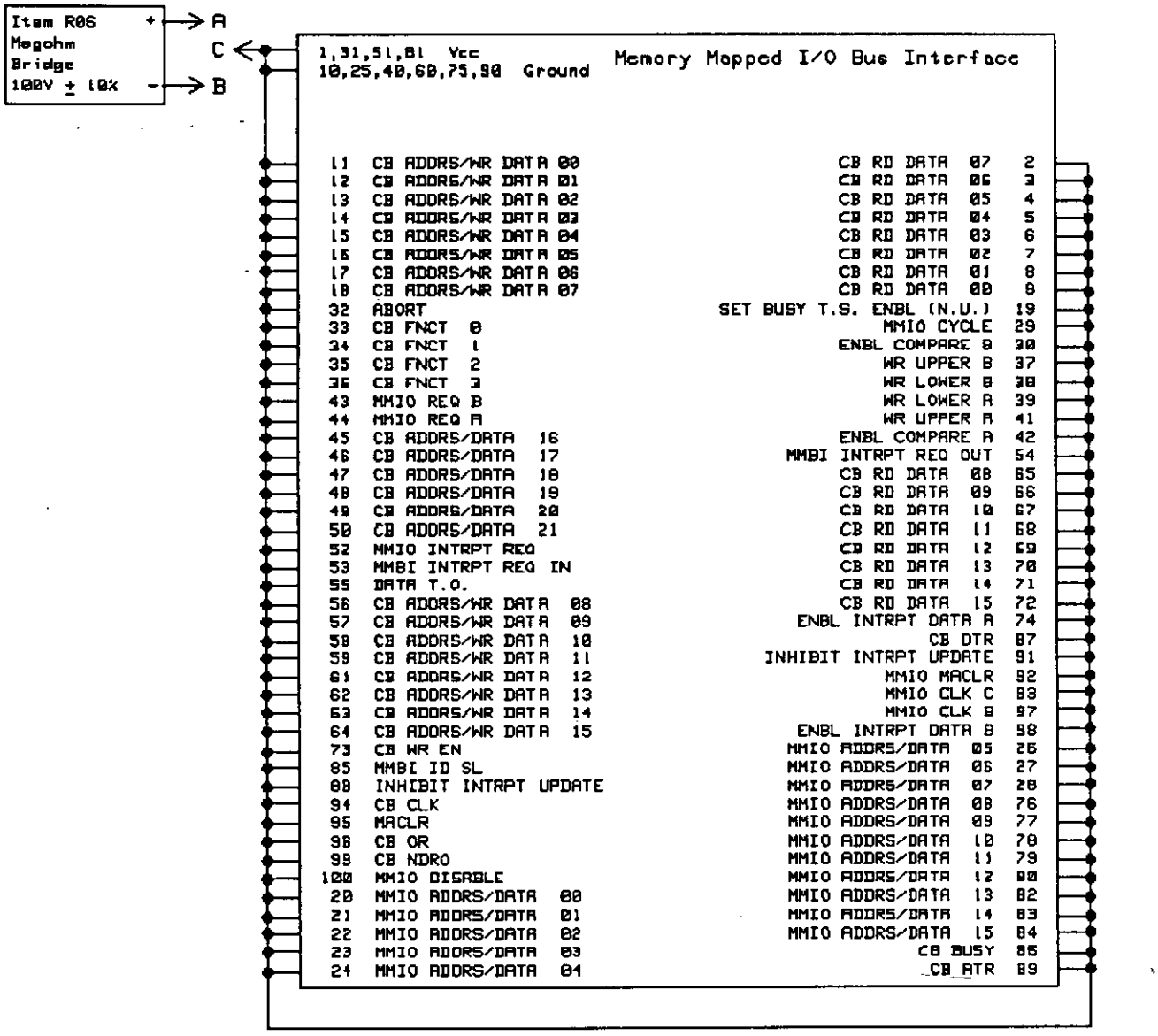


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,51,81 Vcc
10,25,40,60,75,90 Ground

Memory Mapped I/O Bus Interface

11	CB ADDRS/WR DATA 00		CB RD DATA 07	2
12	CB ADDRS/WR DATA 01		CB RD DATA 06	3
13	CB ADDRS/WR DATA 02		CB RD DATA 05	4
14	CB ADDRS/WR DATA 03		CB RD DATA 04	5
15	CB ADDRS/WR DATA 04		CB RD DATA 03	6
16	CB ADDRS/WR DATA 05		CB RD DATA 02	7
17	CB ADDRS/WR DATA 06		CB RD DATA 01	8
18	CB ADDRS/WR DATA 07		CB RD DATA 00	9
32	ABORT		SET BUSY T.S. ENBL (N.U.)	19
33	CB FNCT 0		MMIO CYCLE	29
34	CB FNCT 1		ENBL COMPARE B	30
35	CB FNCT 2		WR UPPER B	37
36	CB FNCT 3		WR LOWER B	38
43	MMIO REQ B		WR LOWER A	39
44	MMIO REQ A		WR UPPER A	41
45	CB ADDRS/DATA 16		ENBL COMPARE A	42
46	CB ADDRS/DATA 17		MMBI INTRPT REQ OUT	54
47	CB ADDRS/DATA 18		CB RD DATA 08	65
48	CB ADDRS/DATA 19		CB RD DATA 09	66
49	CB ADDRS/DATA 20		CB RD DATA 10	67
50	CB ADDRS/DATA 21		CB RD DATA 11	68
52	MMIO INTRPT REQ		CB RD DATA 12	69
53	MMBI INTRPT REQ IN		CB RD DATA 13	70
55	DATA T.O.		CB RD DATA 14	71
56	CB ADDRS/WR DATA 08		CB RD DATA 15	72
57	CB ADDRS/WR DATA 09		ENBL INTRPT DATA A	74
58	CB ADDRS/WR DATA 10		CB DTR	87
59	CB ADDRS/WR DATA 11		INHIBIT INTRPT UPDATE	91
61	CB ADDRS/WR DATA 12		MMIO MACLR	92
62	CB ADDRS/WR DATA 13		MMIO CLK C	93
63	CB ADDRS/WR DATA 14		MMIO CLK B	97
64	CB ADDRS/WR DATA 15		ENBL INTRPT DATA B	98
73	CB WR EN		MMIO ADDRS/DATA 05	26
85	MMBI ID SL		MMIO ADDRS/DATA 06	27
88	INHIBIT INTRPT UPDATE		MMIO ADDRS/DATA 07	28
94	CB CLK		MMIO ADDRS/DATA 08	76
95	MACLR		MMIO ADDRS/DATA 09	77
96	CB OR		MMIO ADDRS/DATA 10	78
98	CB NDRO		MMIO ADDRS/DATA 11	79
100	MMIO DISABLE		MMIO ADDRS/DATA 12	80
20	MMIO ADDRS/DATA 00		MMIO ADDRS/DATA 13	82
21	MMIO ADDRS/DATA 01		MMIO ADDRS/DATA 14	83
22	MMIO ADDRS/DATA 02		MMIO ADDRS/DATA 15	84
23	MMIO ADDRS/DATA 03		CB BUSY	86
24	MMIO ADDRS/DATA 04		CB RTR	89

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.16	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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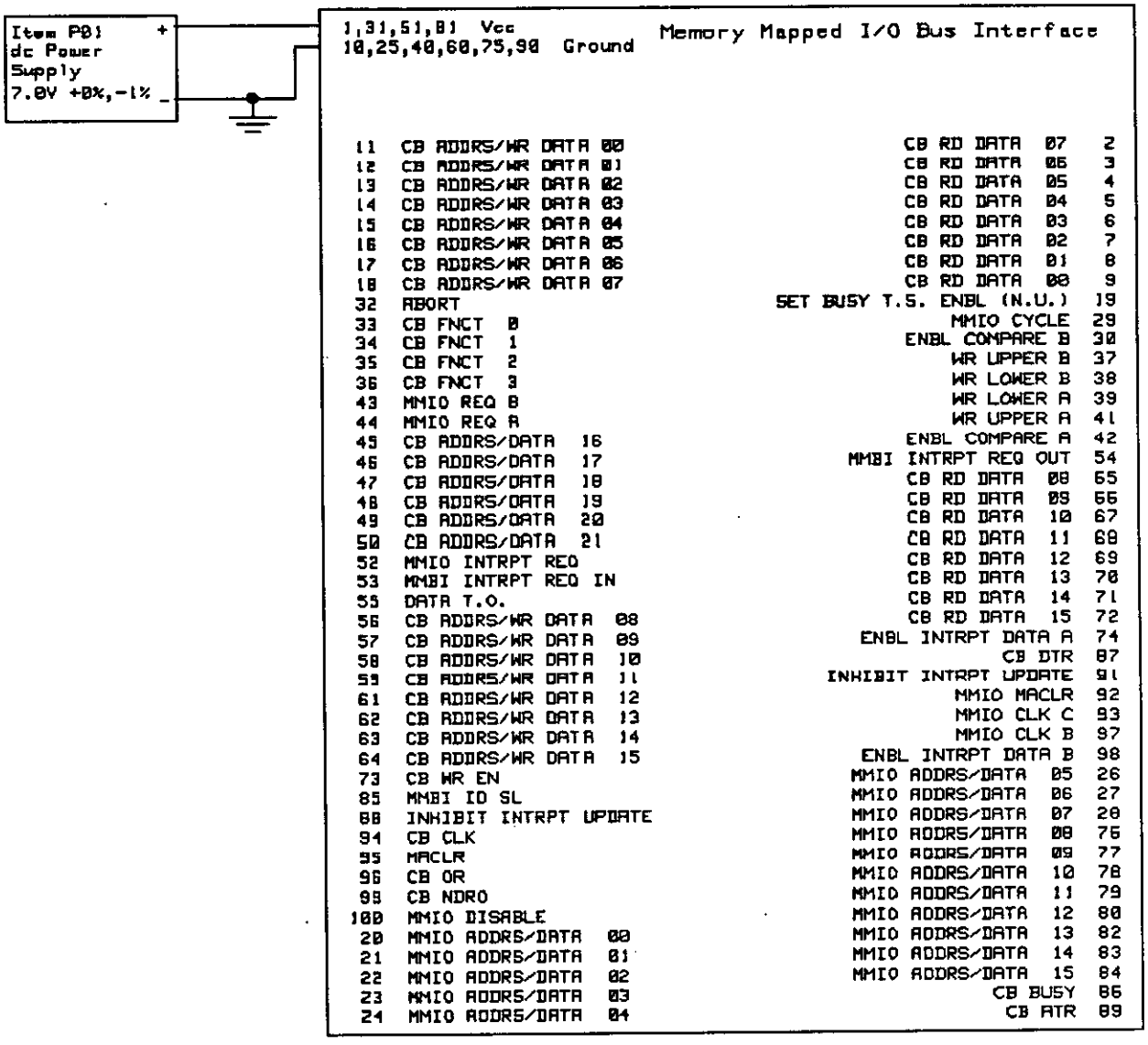


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	CB DATA 00	11	5.5	0.0	0.3	mA	1-3
D2	CB DATA 01	12					
D3	CB DATA 02	13					
D4	CB DATA 03	14	5.5	0.0	0.3	mA	1-3
D5	CB DATA 04	15	5.5	0.0	0.2	mA	1-3
D6	CB DATA 05	16					
D7	CB DATA 06	17					
D8	CB DATA 07	18	5.5	0.0	0.2	mA	1-3
D9	ABORT	32	5.5	0.0	0.1	mA	1-3
D10	CB FNCT 0	33	5.5	0.0	0.2	mA	1-3
D11	CB FNCT 1	34					
D12	CB FNCT 2	35					
D13	CB FNCT 3	36	5.5	0.0	0.2	mA	1-3
D14	REQ B	43	5.5	-0.58	0.68	mA	1-3
D15	REQ A	44	5.5	-0.58	0.68	mA	1-3
D16	CB DATA 16	45	5.5	0.0	0.1	mA	1-3
D17	CB DATA 17	46					
D18	CB DATA 18	47					
D19	CB DATA 19	48					
D20	CB DATA 20	49	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	CB DATA 21	50	5.5	0.0	0.1	mA	1-3
D22	INRP REQ	52	5.5	0.0	0.2	mA	1-3
D23	INRP REQ I	53	5.5	-0.58	1.67	mA	1-3
D24	DATA T.O.	55	5.5	0.0	0.1	mA	1-3
D25	CB DATA 08	56					
D26	CB DATA 09	57					
D27	CB DATA 10	58					
D28	CB DATA 11	59					
D29	CB DATA 12	61	5.5	0.0	0.1	mA	1-3
D30	CB DATA 13	62	5.5	0.0	0.2	mA	1-3
D31	CB DATA 14	63					
D32	CB DATA 15	64	5.5	0.0	0.2	mA	1-3
D33	CB WR EN	73	5.5	0.0	0.1	mA	1-3
D34	MMBI ID SL	85					
D35	INHIBIT UP	88	5.5	0.0	0.1	mA	1-3
D36	CB CLK	94	5.5	0.0	4.0	mA	1-3
D37	MACLR	95	5.5	0.0	0.1	mA	1-3
D38	CB OR	96	5.5	0.0	0.2	mA	1-3
D39	CB NDRO	99	5.5	0.0	0.1	mA	1-3
D40	MMIO DSABL	100	5.5	0.0	0.2	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D40
Power supply current limit	1.8 A	NA	D1-D40
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D40
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D40 D1-D40
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 15 uA ± 10 uA ± 5 uA ± 10 uA ± 63 uA ± 5 uA ± 10 uA ± 112 uA ± 5 uA ± 10 uA ± 5 uA ± 200 uA ± 5 uA ± 10 uA ± 5 uA ± 10 uA	D1-D4 D5-D8 D9 D10-D13 D14,D15 D16-D21 D22 D23 D24-D29 D30-D32 D33-D35 D36 D37 D38 D39 D40

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	RD DATA 07	2	0.0	-150.0	-60.0	mA	1-4
E2	RD DATA 07	2					
E3	RD DATA 06	3					
E4	RD DATA 06	3					
E5	RD DATA 05	4					
E6	RD DATA 05	4					
E7	RD DATA 04	5					
E8	RD DATA 04	5					
E9	RD DATA 03	6					
E10	RD DATA 03	6					
E11	RD DATA 02	7					
E12	RD DATA 02	7					
E13	RD DATA 01	8					
E14	RD DATA 01	8					
E15	RD DATA 00	9					
E16	RD DATA 00	9					
E17	SET BUSY	19					
E18	MM DATA 00	20					
E19	MM DATA 01	21					
E20	MM DATA 02	22	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
E21	MM DATA 03	23	0.0	-150.0	-60.0	mA	1-4
E22	MM DATA 04	24					
E23	MM DATA 05	26					
E24	MM DATA 06	27					
E25	MM DATA 07	28					
E26	COMPARE B	30					
E27	UPPER B	37					
E28	LOWER B	38					
E29	LOWER A	39					
E30	UPPER A	41					
E31	COMPARE A	42					
E32	INRP REQ O	54					
E33	RD DATA 08	65					
E34	RD DATA 08	65					
E35	RD DATA 09	66					
E36	RD DATA 09	66					
E37	RD DATA 10	67					
E38	RD DATA 10	67					
E39	RD DATA 11	68					
E40	RD DATA 11	68	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
E41	RD DATA 12	69	0.0	-150.0	-60.0	mA	1-4
E42	RD DATA 12	69					
E43	RD DATA 13	70					
E44	RD DATA 13	70					
E45	RD DATA 14	71					
E46	RD DATA 14	71					
E47	RD DATA 15	72					
E48	RD DATA 15	72					
E49	EBL DATA A	74					
E50	MM DATA 08	76					
E51	MM DATA 08	76					
E52	MM DATA 09	77					
E53	MM DATA 09	77					
E54	MM DATA 10	78					
E55	MM DATA 10	78					
E56	MM DATA 11	79					
E57	MM DATA 11	79					
E58	MM DATA 12	80					
E59	MM DATA 12	80					
E60	MM DATA 13	82	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E61	MM DATA 13	82	0.0	-150.0	-60.0	mA	1-4
E62	MM DATA 14	83					
E63	MM DATA 14	83					
E64	MM DATA 15	84					
E65	MM DATA 15	84	0.0	-150.0	-60.0	mA	1-4
E66	CB BUSY	86	0.0	-225.0	-100.0	mA	1-4
E67	CB DTR	87					
E68	CB ATR	89					
E69	INH UPDATE	91					
E70	MMIO MACLR	92	0.0	-225.0	-100.0	mA	1-4
E71	MMIO CLK C	93	0.0	-225.0	-50.0	mA	1-4
E72	MMIO CLK B	97	0.0	-225.0	-50.0	mA	1-4
E73	EBL DATA B	98	0.0	-150.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E73
Power supply current limit	1.8 A	NA	E1-E73
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E73
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E73 E1-E73
Output current measurement	 NA NA NA NA	 ± 4.5 mA ± 6.25 mA ± 8.75 mA ± 4.5 mA	 E1-E65 E66-E70 E71,E72 E73

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F1	CB DATA 00	11	2.4	-1.56	0.06	mA	1-3
F2	CB DATA 01	12					
F3	CB DATA 02	13					
F4	CB DATA 03	14	2.4	-1.56	0.06	mA	1-3
F5	CB DATA 04	15	2.4	-1.2	0.04	mA	1-3
F6	CB DATA 05	16					
F7	CB DATA 06	17					
F8	CB DATA 07	18	2.4	-1.2	0.04	mA	1-3
F9	MM DATA 00	20	2.4	-0.65	0.07	mA	1-3
F10	MM DATA 01	21					
F11	MM DATA 02	22					
F12	MM DATA 03	23					
F13	MM DATA 04	24					
F14	MM DATA 05	26					
F15	MM DATA 06	27					
F16	MM DATA 07	28	2.4	-0.65	0.07	mA	1-3
F17	ABORT	32	2.4	-0.6	0.02	mA	1-3
F18	CB FNCT 0	33	2.4	-0.96	0.04	mA	1-3
F19	CB FNCT 1	34					
F20	CB FNCT 2	35	2.4	-0.96	0.04	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	CB FNCT 3	36	2.4	-0.96	0.04	mA	1-3
F22	REQ B	43	2.4	-27.7	-14.3	mA	1-3
F23	REQ A	44	2.4	-27.7	-14.3	mA	1-3
F24	CB DATA 16	45	2.4	-0.6	0.02	mA	1-3
F25	CB DATA 17	46					
F26	CB DATA 18	47					
F27	CB DATA 19	48					
F28	CB DATA 20	49					
F29	CB DATA 21	50	2.4	-0.6	0.02	mA	1-3
F30	INRP REQ	52	2.4	-1.32	0.06	mA	1-3
F31	INRP REQ I	53	2.4	-28.1	-14.3	mA	1-3
F32	DATA T.O.	55	2.4	-0.2	0.02	mA	1-3
F33	CB DATA 08	56	2.4	-0.6	0.02	mA	1-3
F34	CB DATA 09	57					
F35	CB DATA 10	58					
F36	CB DATA 11	59					
F37	CB DATA 12	61	2.4	-0.6	0.02	mA	1-3
F38	CB DATA 13	62	2.4	-1.2	0.04	mA	1-3
F39	CB DATA 14	63					
F40	CB DATA 15	64	2.4	-1.2	0.04	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	CB WR EN	73	2.4	-0.6	0.02	mA	1-3
F42	MM DATA 08	76	2.4	-1.3	0.14	mA	1-3
F43	MM DATA 09	77					
F44	MM DATA 10	78					
F45	MM DATA 11	79					
F46	MM DATA 12	80					
F47	MM DATA 13	82					
F48	MM DATA 14	83					
F49	MM DATA 15	84	2.4	-1.3	0.14	mA	1-3
F50	MMBI ID SL	85	2.4	-0.2	0.02	mA	1-3
F51	CB BUSY	86	2.4	-0.65	0.07	mA	1-3
F52	INHIBIT UP	88	2.4	-1.6	0.02	mA	1-3
F53	CB ATR	89	2.4	-0.65	0.07	mA	1-3
F54	CB CLK	94	2.4	-1.6	0.2	mA	1-3
F55	MACLR	95	2.4	-1.6	0.02	mA	1-3
F56	CB OR	96	2.4	-1.2	0.04	mA	1-3
F57	CB NDRO	99	2.4	-0.6	0.02	mA	1-3
F58	MMIO DSABL	100	2.4	-0.8	0.04	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F58
Power supply current limit	1.8 A	NA	F1-F58
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F58
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F58
low, (VL)	0.0 V	± 100 mV	F1-F58
Input current measurement	NA	± 81 uA	F1-F4
	NA	± 62 uA	F5-F8
	NA	± 36 uA	F9-F16
	NA	± 31 uA	F17
	NA	± 50 uA	F18-F21
	NA	± 670 uA	F22,F23
	NA	± 31 uA	F24-F29
	NA	± 69 uA	F30
	NA	± 690 uA	F31
	NA	± 11 uA	F32
	NA	± 31 uA	F33-F37
	NA	± 62 uA	F38-F40
	NA	± 31 uA	F41
	NA	± 72 uA	F42-F49
	NA	± 11 uA	F50
	NA	± 36 uA	F51
	NA	± 81 uA	F52
	NA	± 36 uA	F53
	NA	± 90 uA	F54
	NA	± 81 uA	F55

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA NA NA	± 62 uA ± 31 uA ± 42 uA	F56 F57 F58

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	CB DATA 00	11	0.4	-1.56	0.06	mA	1-3
G2	CB DATA 01	12					
G3	CB DATA 02	13					
G4	CB DATA 03	14	0.4	-1.56	0.06	mA	1-3
G5	CB DATA 04	15	0.4	-1.2	0.04	mA	1-3
G6	CB DATA 05	16					
G7	CB DATA 06	17					
G8	CB DATA 07	18	0.4	-1.2	0.04	mA	1-3
G9	MM DATA 00	20	0.4	-0.65	0.07	mA	1-3
G10	MM DATA 01	21					
G11	MM DATA 02	22					
G12	MM DATA 03	23					
G13	MM DATA 04	24					
G14	MM DATA 05	26					
G15	MM DATA 06	27					
G16	MM DATA 07	28	0.4	-0.65	0.07	mA	1-3
G17	ABORT	32	0.4	-0.6	0.02	mA	1-3
G18	CB FNCT 0	33	0.4	-0.96	0.04	mA	1-3
G19	CB FNCT 1	34					
G20	CB FNCT 2	35	0.4	-0.96	0.04	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	CB FNCT 3	36	0.4	-0.96	0.04	mA	1-3
G22	REQ B	43	0.4	-27.7	-14.3	mA	1-3
G23	REQ A	44	0.4	-27.7	-14.3	mA	1-3
G24	CB DATA 16	45	0.4	-0.6	0.02	mA	1-3
G25	CB DATA 17	46					
G26	CB DATA 18	47					
G27	CB DATA 19	48					
G28	CB DATA 20	49					
G29	CB DATA 21	50	0.4	-0.6	0.02	mA	1-3
G30	INRP REQ	52	0.4	-1.32	0.06	mA	1-3
G31	INRP REQ I	53	0.4	-28.1	-14.3	mA	1-3
G32	DATA T.O.	55	0.4	-0.2	0.02	mA	1-3
G33	CB DATA 08	56	0.4	-0.6	0.02	mA	1-3
G34	CB DATA 09	57					
G35	CB DATA 10	58					
G36	CB DATA 11	59					
G37	CB DATA 12	61	0.4	-0.6	0.02	mA	1-3
G38	CB DATA 13	62	0.4	-1.2	0.04	mA	1-3
G39	CB DATA 14	63					
G40	CB DATA 15	64	0.4	-1.2	0.04	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	CB WR EN	73	0.4	-0.6	0.02	mA	1-3
G42	MM DATA 08	76	0.4	-1.3	0.14	mA	1-3
G43	MM DATA 09	77					
G44	MM DATA 10	78					
G45	MM DATA 11	79					
G46	MM DATA 12	80					
G47	MM DATA 13	82					
G48	MM DATA 14	83					
G49	MM DATA 15	84	0.4	-1.3	0.14	mA	1-3
G50	MMBI ID SL	85	0.4	-0.2	0.02	mA	1-3
G51	CB BUSY	86	0.4	-0.65	0.07	mA	1-3
G52	INHIBIT UP	88	0.4	-1.6	0.02	mA	1-3
G53	CB ATR	89	0.4	-0.65	0.07	mA	1-3
G54	CB CLK	94	0.4	-1.6	0.2	mA	1-3
G55	MACLR	95	0.4	-1.6	0.02	mA	1-3
G56	CB OR	96	0.4	-1.2	0.04	mA	1-3
G57	CB NDRO	99	0.4	-0.6	0.02	mA	1-3
G58	MMIO DSABL	100	0.4	-0.8	0.04	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G58
Power supply current limit	1.8 A	NA	G1-G58
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G58
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G58 G1-G58
Input current measurement	NA	± 81 uA	G1-G4
	NA	± 62 uA	G5-G8
	NA	± 36 uA	G9-G16
	NA	± 31 uA	G17
	NA	± 50 uA	G18-G21
	NA	± 670 uA	G22,G23
	NA	± 31 uA	G24-G29
	NA	± 69 uA	G30
	NA	± 690 uA	G31
	NA	± 11 uA	G32
	NA	± 31 uA	G33-G37
	NA	± 62 uA	G38-G40
	NA	± 31 uA	G41
	NA	± 72 uA	G42-G49
	NA	± 11 uA	G50
	NA	± 36 uA	G51
	NA	± 81 uA	G52
	NA	± 36 uA	G53
	NA	± 90 uA	G54
	NA	± 81 uA	G55

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA NA NA	± 62 uA ± 31 uA ± 42 uA	G56 G57 G58

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	RD DATA 07	2	-0.4	2.4	4.5	V	1-4
H2	RD DATA 07	2					
H3	RD DATA 06	3					
H4	RD DATA 06	3					
H5	RD DATA 05	4					
H6	RD DATA 05	4					
H7	RD DATA 04	5					
H8	RD DATA 04	5					
H9	RD DATA 03	6					
H10	RD DATA 03	6					
H11	RD DATA 02	7					
H12	RD DATA 02	7					
H13	RD DATA 01	8					
H14	RD DATA 01	8					
H15	RD DATA 00	9					
H16	RD DATA 00	9					
H17	SET BUSY	19					
H18	MM DATA 00	20					
H19	MM DATA 01	21					
H20	MM DATA 02	22	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	MM DATA 03	23	-0.4	2.4	4.5	V	1-4
H22	MM DATA 04	24					
H23	MM DATA 05	26					
H24	MM DATA 06	27					
H25	MM DATA 07	28					
H26	COMPARE B	30					
H27	UPPER B	37					
H28	LOWER B	38					
H29	LOWER A	39					
H30	UPPER A	41					
H31	COMPARE A	42					
H32	INRP REQ O	54					
H33	RD DATA 08	65					
H34	RD DATA 08	65					
H35	RD DATA 09	66					
H36	RD DATA 09	66					
H37	RD DATA 10	67					
H38	RD DATA 10	67					
H39	RD DATA 11	68					
H40	RD DATA 11	68	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H41	RD DATA 12	69	-0.4	2.4	4.5	V	1-4
H42	RD DATA 12	69					
H43	RD DATA 13	70					
H44	RD DATA 13	70					
H45	RD DATA 14	71					
H46	RD DATA 14	71					
H47	RD DATA 15	72					
H48	RD DATA 15	72					
H49	EBL DATA A	74					
H50	MM DATA 08	76					
H51	MM DATA 08	76					
H52	MM DATA 09	77					
H53	MM DATA 09	77					
H54	MM DATA 10	78					
H55	MM DATA 10	78					
H56	MM DATA 11	79					
H57	MM DATA 11	79					
H58	MM DATA 12	80					
H59	MM DATA 12	80					
H60	MM DATA 13	82	-0.4	2.4	4.5	V	1-4

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Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H61	MM DATA 13	82	-0.4	2.4	4.5	V	1-4
H62	MM DATA 14	83					
H63	MM DATA 14	83					
H64	MM DATA 15	84					
H65	MM DATA 15	84					
H66	CB BUSY	86					
H67	CB DTR	87					
H68	CB ATR	89					
H69	INH UPDATE	91					
H70	MMIO MACLR	92					
H71	MMIO CLK C	93					
H72	MMIO CLK B	97					
H73	EBL DATA B	98	-0.4	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 20261O-20T18:27Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H73
Power supply current limit	1.8 A	NA	H1-H73
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H73
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H73 H1-H73
Output voltage measurement	NA	± 105 mV	H1-H73

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

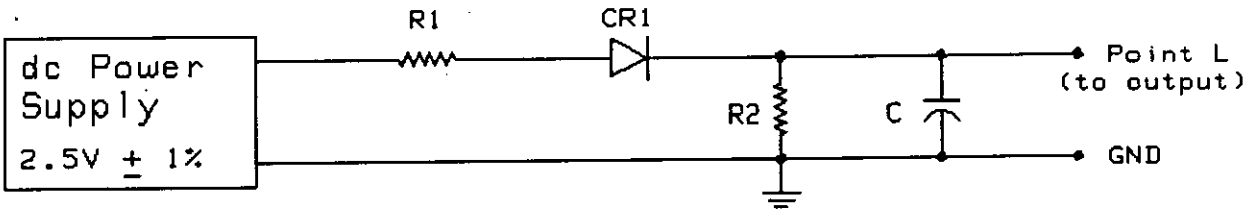
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,3,4,5,6,7,8,9, 19,20,21,22,23, 24,26,27,28,30, 37,38,39,41,42, 54,65,66,67,68, 69,70,71,72,74, 76,77,78,79,80, 82,83,84,86,87, 89,91,92,93,97, 98	6200	6.0	0.5
29	NA	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	RD DATA 07	2	6.0	0.0	0.5	V	1-4
J2	RD DATA 07	2					
J3	RD DATA 06	3					
J4	RD DATA 06	3					
J5	RD DATA 05	4					
J6	RD DATA 05	4					
J7	RD DATA 04	5					
J8	RD DATA 04	5					
J9	RD DATA 03	6					
J10	RD DATA 03	6					
J11	RD DATA 02	7					
J12	RD DATA 02	7					
J13	RD DATA 01	8					
J14	RD DATA 01	8					
J15	RD DATA 00	9					
J16	RD DATA 00	9					
J17	SET BUSY	19					
J18	MM DATA 00	20					
J19	MM DATA 01	21					
J20	MM DATA 02	22	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	MM DATA 03	23	6.0	0.0	0.5	V	1-4
J22	MM DATA 04	24					
J23	MM DATA 05	26					
J24	MM DATA 06	27					
J25	MM DATA 07	28					
J26	MMIO CYCLE	29					
J27	COMPARE B	30					
J28	UPPER B	37					
J29	LOWER B	38					
J30	LOWER A	39					
J31	UPPER A	41					
J32	COMPARE A	42					
J33	INRP REQ O	54					
J34	RD DATA 08	65					
J35	RD DATA 09	66					
J36	RD DATA 10	67					
J37	RD DATA 11	68					
J38	RD DATA 12	69					
J39	RD DATA 13	70					
J40	RD DATA 14	71	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	RD DATA 15	72	6.0	0.0	0.5	V	1-4
J42	EBL DATA A	74					
J43	MM DATA 08	76					
J44	MM DATA 08	76					
J45	MM DATA 09	77					
J46	MM DATA 09	77					
J47	MM DATA 10	78					
J48	MM DATA 10	78					
J49	MM DATA 11	79					
J50	MM DATA 11	79					
J51	MM DATA 12	80					
J52	MM DATA 12	80					
J53	MM DATA 13	82					
J54	MM DATA 13	82					
J55	MM DATA 14	83					
J56	MM DATA 14	83					
J57	MM DATA 15	84					
J58	MM DATA 15	84					
J59	CB BUSY	86					
J60	CB DTR	87	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J61	CB ATR	89	6.0	0.0	0.5	V	1-4
J62	INH UPDATE	91					
J63	MMIO MACLR	92					
J64	MMIO CLK C	93					
J65	MMIO CLK B	97					
J66	EBL DATA B	98	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J66
Power supply current limit	1.8 A	NA	J1-J66
Forcing current, (Iol)	6.0 mA	± 600 uA	J1-J66
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J66 J1-J66
Output voltage measurement	NA	± 25 mV	J1-J66

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
High level							
K1	RD DATA 07	2	2.4	-0.1	0.1	mA	1-3
K2	RD DATA 06	3					
K3	RD DATA 05	4					
K4	RD DATA 04	5					
K5	RD DATA 03	6					
K6	RD DATA 02	7					
K7	RD DATA 01	8					
K8	RD DATA 00	9	2.4	-0.1	0.1	mA	1-3
K9	MMIO CYCLE	29	5.5	0.0	0.25	mA	1-3
K10	RD DATA 08	65	2.4	-0.1	0.1	mA	1-3
K11	RD DATA 09	66					
K12	RD DATA 10	67					
K13	RD DATA 11	68					
K14	RD DATA 12	69					
K15	RD DATA 13	70					
K16	RD DATA 14	71					
K17	RD DATA 15	72	2.4	-0.1	0.1	mA	1-3
K18	CB DTR	87	2.4	-0.05	0.05	mA	1-3

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TABLE XI. Output leakage current, high and low level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
Low level							
K19	RD DATA 07	2	0.4	-0.1	0.1	mA	1-3
K20	RD DATA 06	3					
K21	RD DATA 05	4					
K22	RD DATA 04	5					
K23	RD DATA 03	6					
K24	RD DATA 02	7					
K25	RD DATA 01	8					
K26	RD DATA 00	9					
K27	RD DATA 08	65					
K28	RD DATA 09	66					
K29	RD DATA 10	67					
K30	RD DATA 11	68					
K31	RD DATA 12	69					
K32	RD DATA 13	70					
K33	RD DATA 14	71					
K34	RD DATA 15	72	0.4	-0.1	0.1	mA	1-3
K35	CB DTR	87	0.4	-0.05	0.05	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K35
Power supply current limit	1.8 A	NA	K1-K35
Forcing voltage,			
(Voh)	2.4 V	± 50 mV	K1-K8
(Voh)	5.5 V	± 50 mV	K9
(Voh)	2.4 V	± 50 mV	K10-K18
(Vol)	0.4 V	± 50 mV	K19-K35
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	K1-K35
low, (VL)	0.0 V	± 100 mV	K1-K35
Output current measurement	NA	± 10.0 uA	K1-K8
	NA	± 12.5 uA	K9
	NA	± 10.0 uA	K10-K17
	NA	± 5.0 uA	K18
	NA	± 10.0 uA	K19-K34
	NA	± 5.0 uA	K35

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	CB DATA 00	11	-18.0	-1.2	0.0	V	1-3
L2	CB DATA 01	12					
L3	CB DATA 02	13					
L4	CB DATA 03	14					
L5	CB DATA 04	15					
L6	CB DATA 05	16					
L7	CB DATA 06	17					
L8	CB DATA 07	18					
L9	ABORT	32					
L10	CB FNCT 0	33					
L11	CB FNCT 1	34					
L12	CB FNCT 2	35					
L13	CB FNCT 3	36					
L14	CB DATA 16	45					
L15	CB DATA 17	46					
L16	CB DATA 18	47					
L17	CB DATA 19	48					
L18	CB DATA 20	49					
L19	CB DATA 21	50					
L20	INRP REQ	52	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	DATA T.O.	55	-18.0	-1.5	0.0	V	1-3
L22	CB DATA 08	56	-18.0	-1.2	0.0	V	1-3
L23	CB DATA 09	57					
L24	CB DATA 10	58					
L25	CB DATA 11	59					
L26	CB DATA 12	61					
L27	CB DATA 13	62					
L28	CB DATA 14	63					
L29	CB DATA 15	64					
L30	CB WR EN	73	-18.0	-1.2	0.0	V	1-3
L31	MMBI ID SL	85	-18.0	-1.5	0.0	V	1-3
L32	INHIBIT UP	88	-18.0	-1.2	0.0	V	1-3
L33	CB CLK	94					
L34	MACLR	95					
L35	CB OR	96					
L36	CB NDRO	99					
L37	MMIO DSABL	100	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L37
Power supply current limit	NA	NA	L1-L37
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L37
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L37 L1-L37
Input voltage measurement	NA NA NA NA NA	± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV	L1-L20 L21 L22-L30 L31 L32-L37

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	7.1	13.2	uF	1-4
M2	PSFC	51	7.1	13.3	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 305 nF ± 310 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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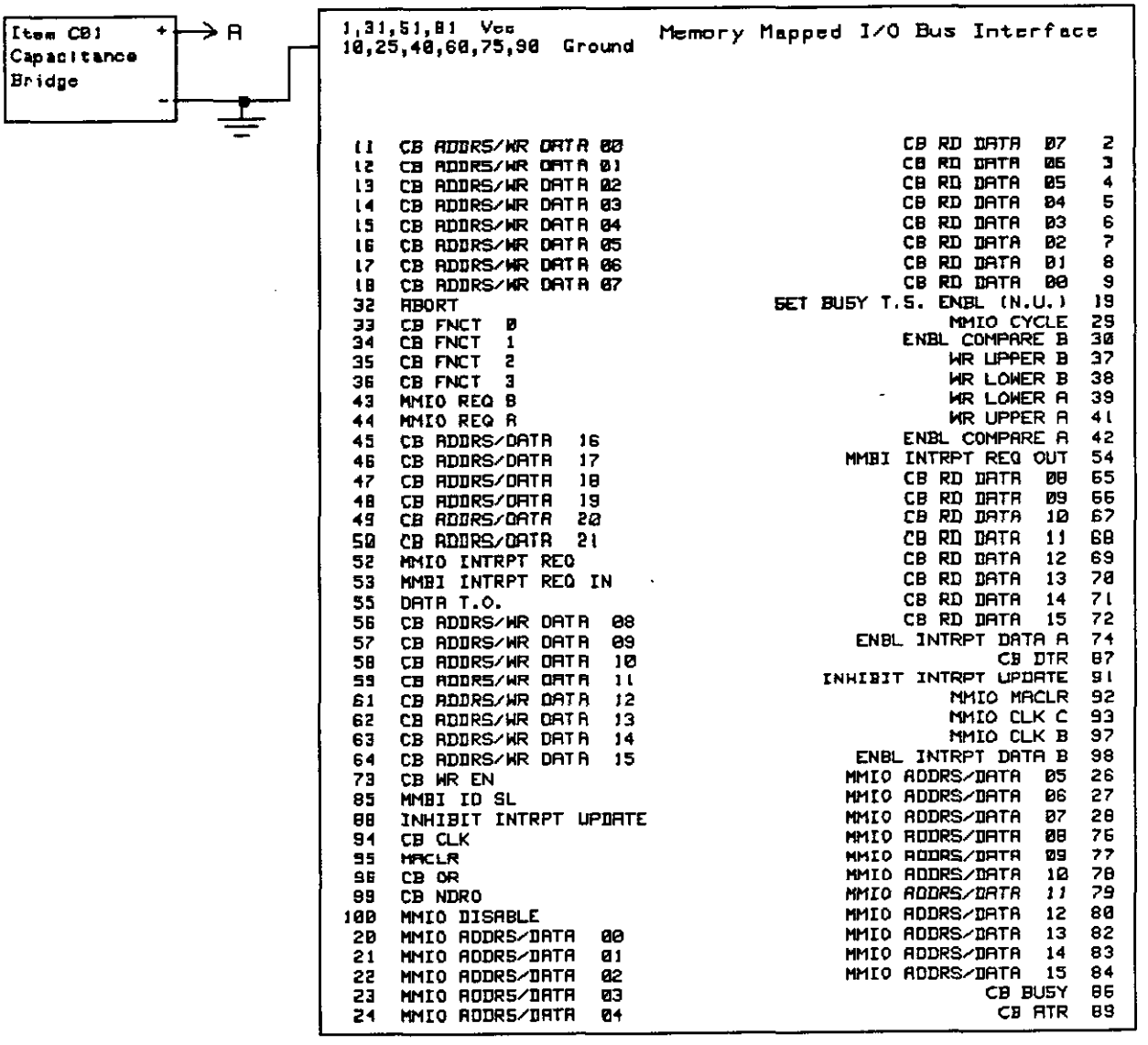


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	500	1200	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.8 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 35 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power suply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.8 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.8 A	NA	Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage			
high, (VH)	2.4 V	± 100 mV	Q1-Q4
low, (VL)	0.4 V	± 100 mV	Q1-Q4
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

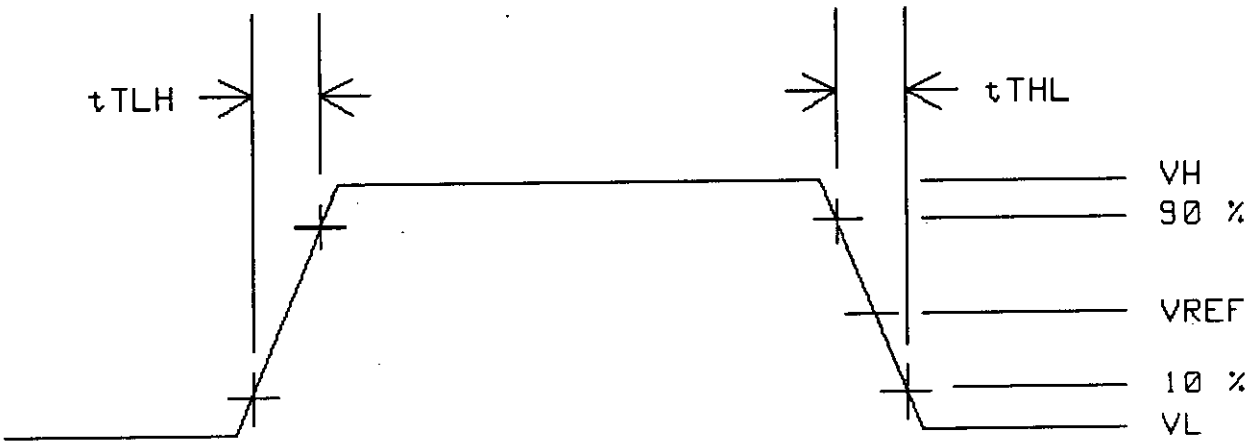
(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

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NOTES: - Continued.

4. I/O waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CB CLK to COMPARE B	-	-	-	36	36	36	nS	11
1	tpd	CB CLK to COMPARE A	-	-	-	36	36	36	nS	11
2	tpd	REQ B to MMIO CYCLE	-	-	-	27	27	27	nS	11
3	tpd	CB CLK to MMIO CYCLE	-	-	-	64	64	64	nS	11
4	tpd	CB CLK to SET BUSY	-	-	-	36	36	36	nS	11
5	tpd	CB CLK to CB BUSY	-	-	-	38	38	38	nS	11
6	tpd	CB CLK to CB DTR	-	-	-	38	38	38	nS	11
7	tpd	CB CLK to RD DATA 00	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 01	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 02	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 03								

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
7	tpd	CB CLK to RD DATA 04	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 05	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 06	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 07	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 08	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 09	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 10	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 11	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 12	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 13	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 14	-	-	-	59	59	59	nS	11
7	tpd	CB CLK to RD DATA 15	-	-	-	59	59	59	nS	11
8	tpd	CB FNCT 2 to LOWER A	-	-	-	27	27	27	nS	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	CB FNCT 2 to LOWER B	-	-	-	27	27	27	nS	11
8	tpd	CB FNCT 2 to UPPER A	-	-	-	27	27	27	nS	11
8	tpd	CB FNCT 2 to UPPER B	-	-	-	27	27	27	nS	11
9	tpd	CB CLK to LOWER A	-	-	-	47	47	47	nS	11
9	tpd	CB CLK to LOWER B	-	-	-	47	47	47	nS	11
9	tpd	CB CLK to UPPER A	-	-	-	47	47	47	nS	11
9	tpd	CB CLK to UPPER B	-	-	-	47	47	47	nS	11
10	tpd	CB DATA 00 to MM DATA 00	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 01 to MM DATA 01	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 02 to MM DATA 02	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 03 to MM DATA 03	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 04 to MM DATA 04	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 05 to MM DATA 05	-	-	-	16	16	16	nS	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
10	tpd	CB DATA 06 to MM DATA 06	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 07 to MM DATA 07	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 08 to MM DATA 08	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 09 to MM DATA 09	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 10 to MM DATA 10	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 11 to MM DATA 11	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 12 to MM DATA 12	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 13 to MM DATA 13	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 14 to MM DATA 14	-	-	-	16	16	16	nS	11
10	tpd	CB DATA 15 to MM DATA 15	-	-	-	16	16	16	nS	11
11	tpd	CB CLK to MM DATA 00	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 01	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 02	-	-	-	65	65	65	nS	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	CB CLK to MM DATA 03	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 04	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 05	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 06	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 07	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 08	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 09	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 10	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 11	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 12	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 13	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 14	-	-	-	65	65	65	nS	11
11	tpd	CB CLK to MM DATA 15	-	-	-	65	65	65	nS	11

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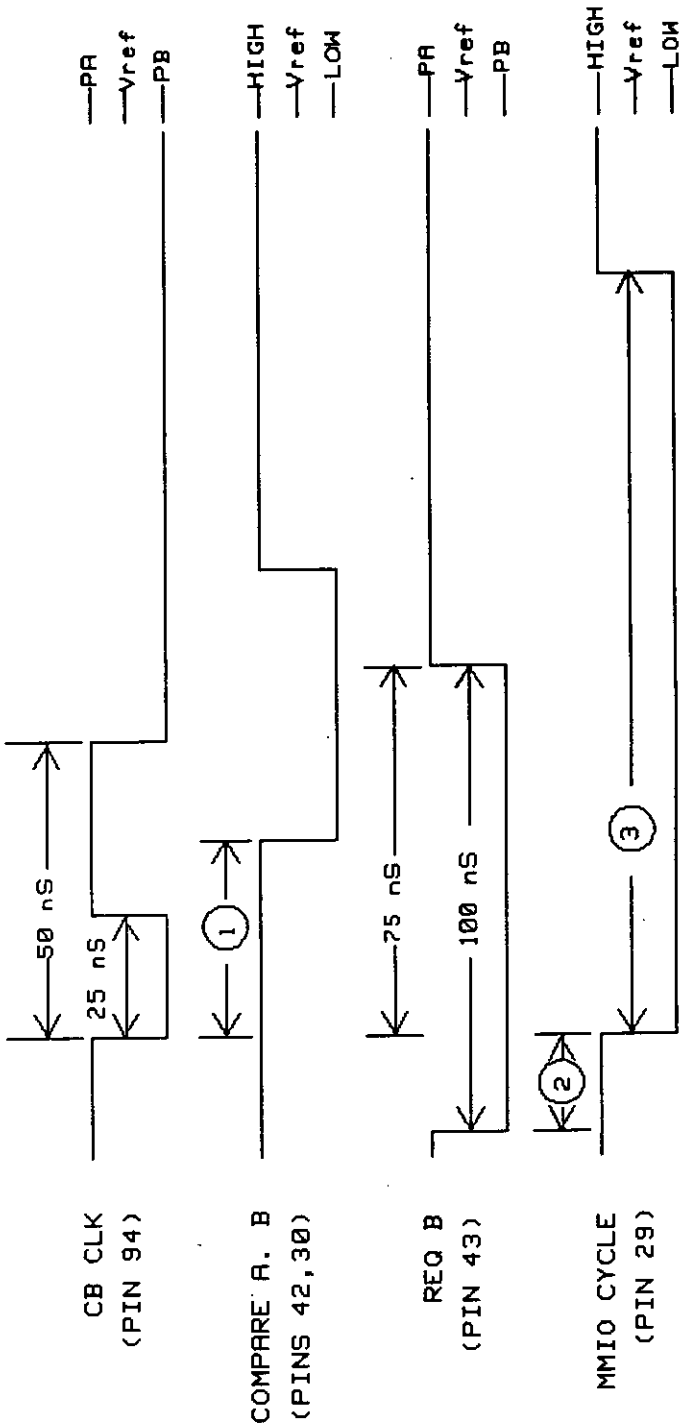


FIGURE 11a. Dynamic waveforms for table XVII.

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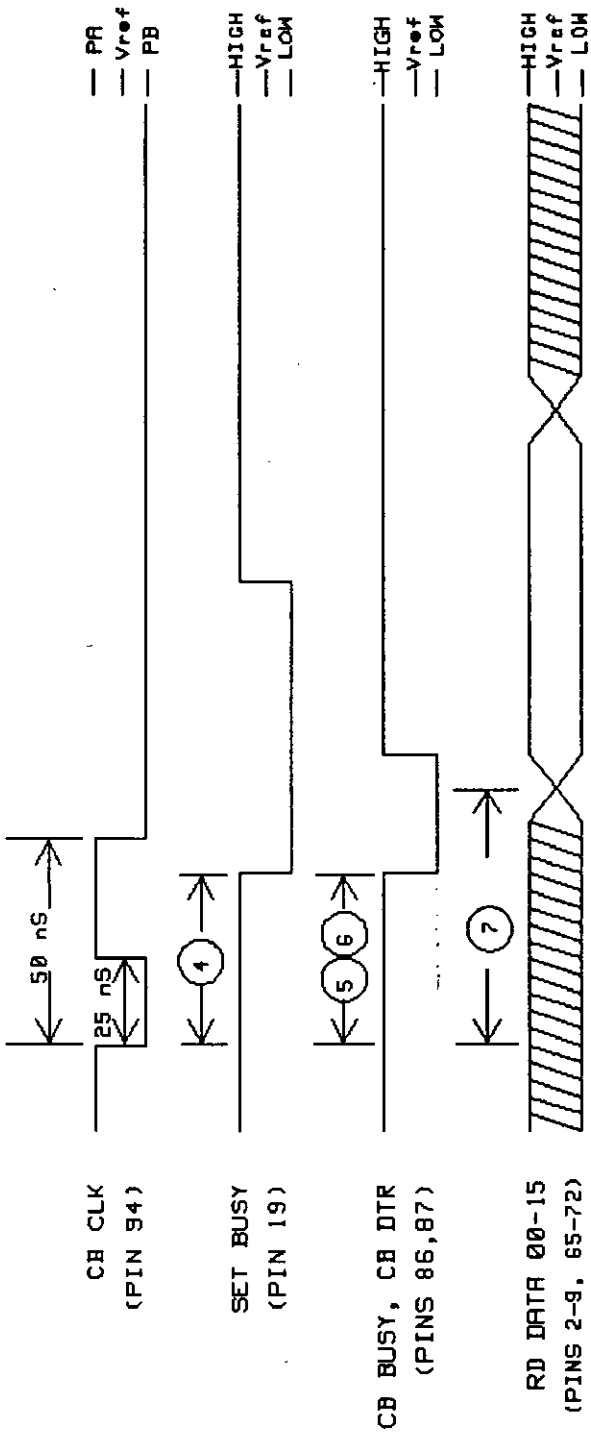


FIGURE 11b. Dynamic waveforms for table XVII.

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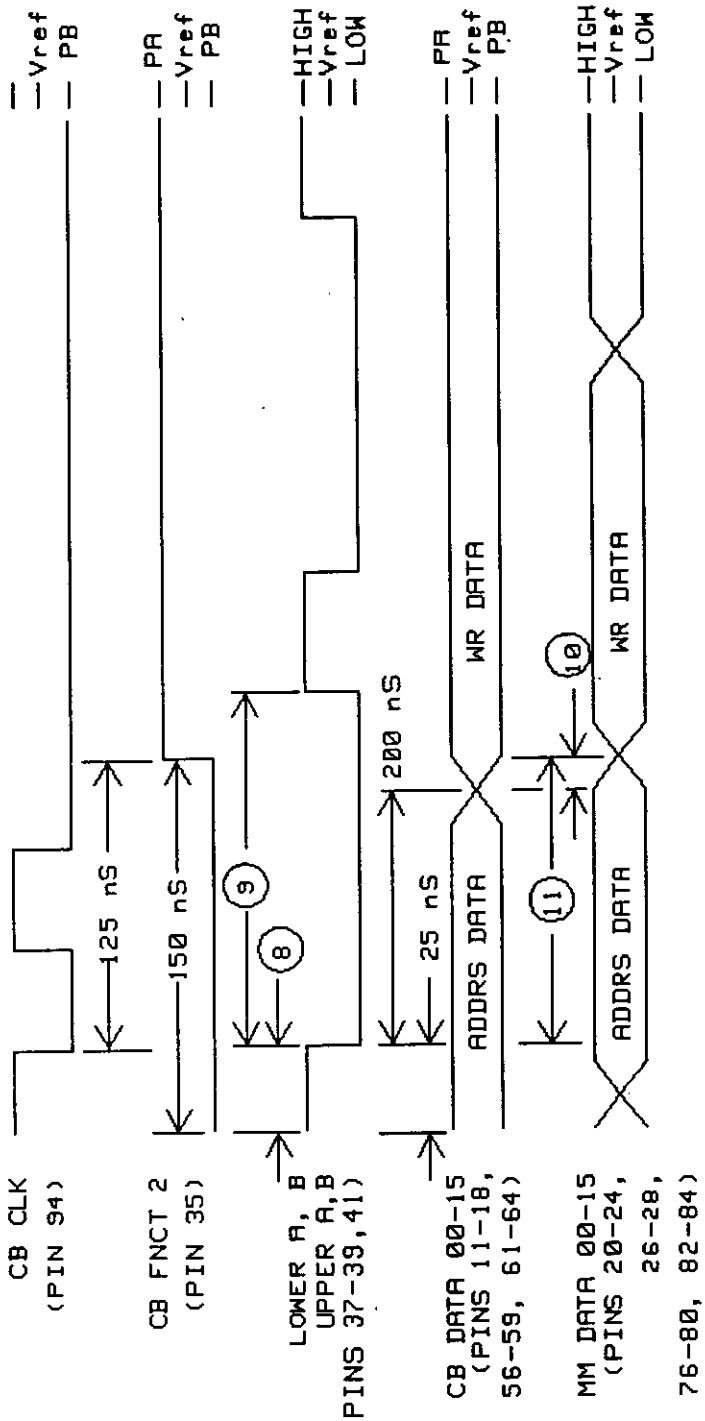


FIGURE 11c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	CB DATA 00	11	6	125	pF	1-4
R2	CB DATA 01	12	6	100	pF	1-4
R3	CB DATA 02	13	6	100	pF	1-4
R4	CB DATA 03	14	6	125	pF	1-4
R5	CB DATA 04	15	4	75	pF	1-4
R6	CB DATA 05	16				
R7	CB DATA 06	17	4	75	pF	1-4
R8	CB DATA 07	18	4	100	pF	1-4
R9	ABORT	32	2	75	pF	1-4
R10	CB FNCT 0	33	4	75	pF	1-4
R11	CB FNCT 1	34				
R12	CB FNCT 2	35	4	75	pF	1-4
R13	CB FNCT 3	36	4	100	pF	1-4
R14	REQ B	43				
R15	REQ A	44	4	100	pF	1-4
R16	CB DATA 16	45	2	75	pF	1-4
R17	CB DATA 17	46	2	50	pF	1-4
R18	CB DATA 18	47	2	75	pF	1-4
R19	CB DATA 19	48				
R20	CB DATA 20	49	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	CB DATA 21	50	2	75	pF	1-4
R22	INRP REQ	52	2	100	pF	1-4
R23	INRP REQ I	53	6	125	pF	1-4
R24	DATA T.O.	55	2	100	pF	1-4
R25	CB DATA 08	56	2	75	pF	1-4
R26	CB DATA 09	57				
R27	CB DATA 10	58				
R28	CB DATA 11	59				
R29	CB DATA 12	61	2	75	pF	1-4
R30	CB DATA 13	62	2	125	pF	1-4
R31	CB DATA 14	63				
R32	CB DATA 15	64	2	125	pF	1-4
R33	CB WR EN	73	2	75	pF	1-4
R34	MMBI ID SL	85				
R35	INHIBIT UP	88	2	75	pF	1-4
R36	CB CLK	94	8	100	pF	1-4
R37	MACLR	95	2	75	pF	1-4
R38	CB OR	96	4	75	pF	1-4
R39	CB NDRO	99	2	75	pF	1-4
R40	MMIO DSABL	100	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Bidirectionals						
R41	MM DATA 00	20	4	75	pF	1-4
R42	MM DATA 01	21				
R43	MM DATA 02	22				
R44	MM DATA 03	23				
R45	MM DATA 04	24				
R46	MM DATA 05	26				
R47	MM DATA 06	27				
R48	MM DATA 07	28	4	75	pF	1-4
R49	MM DATA 08	76	2	150	pF	1-4
R50	MM DATA 09	77	2	125	pF	1-4
R51	MM DATA 10	78	2	125	pF	1-4
R52	MM DATA 11	79	2	150	pF	1-4
R53	MM DATA 12	80	2	125	pF	1-4
R54	MM DATA 13	82	2	150	pF	1-4
R55	MM DATA 14	83	2	125	pF	1-4
R56	MM DATA 15	84	2	125	pF	1-4
R57	CB BUSY	86	4	125	pF	1-4
R58	CB ATR	89	4	75	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R58
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R58
Capacitance measurement	NA	± 5.95 pF	R1
	NA	± 4.70 pF	R2,R3
	NA	± 5.95 pF	R4
	NA	± 3.55 pF	R5-R7
	NA	± 4.80 pF	R8
	NA	± 3.65 pF	R9
	NA	± 3.55 pF	R10-R12
	NA	± 4.80 pF	R13-R15
	NA	± 3.65 pF	R16
	NA	± 2.40 pF	R17
	NA	± 3.65 pF	R18-R21
	NA	± 4.90 pF	R22
	NA	± 5.95 pF	R23
	NA	± 4.90 pF	R24
	NA	± 3.65 pF	R25-R29
	NA	± 6.15 pF	R30-R32
	NA	± 3.65 pF	R33-R35
	NA	± 4.60 pF	R36
	NA	± 3.65 pF	R37
	NA	± 3.55 pF	R38
	NA	± 3.65 pF	R39,R40
	NA	± 3.55 pF	R41-R48
	NA	± 7.40 pF	R49
	NA	± 6.15 pF	R50,R51
	NA	± 7.40 pF	R52
	NA	± 6.15 pF	R53
	NA	± 7.40 pF	R54
	NA	± 6.15 pF	R55,R56
	NA	± 6.05 pF	R57
	NA	± 3.55 pF	R58

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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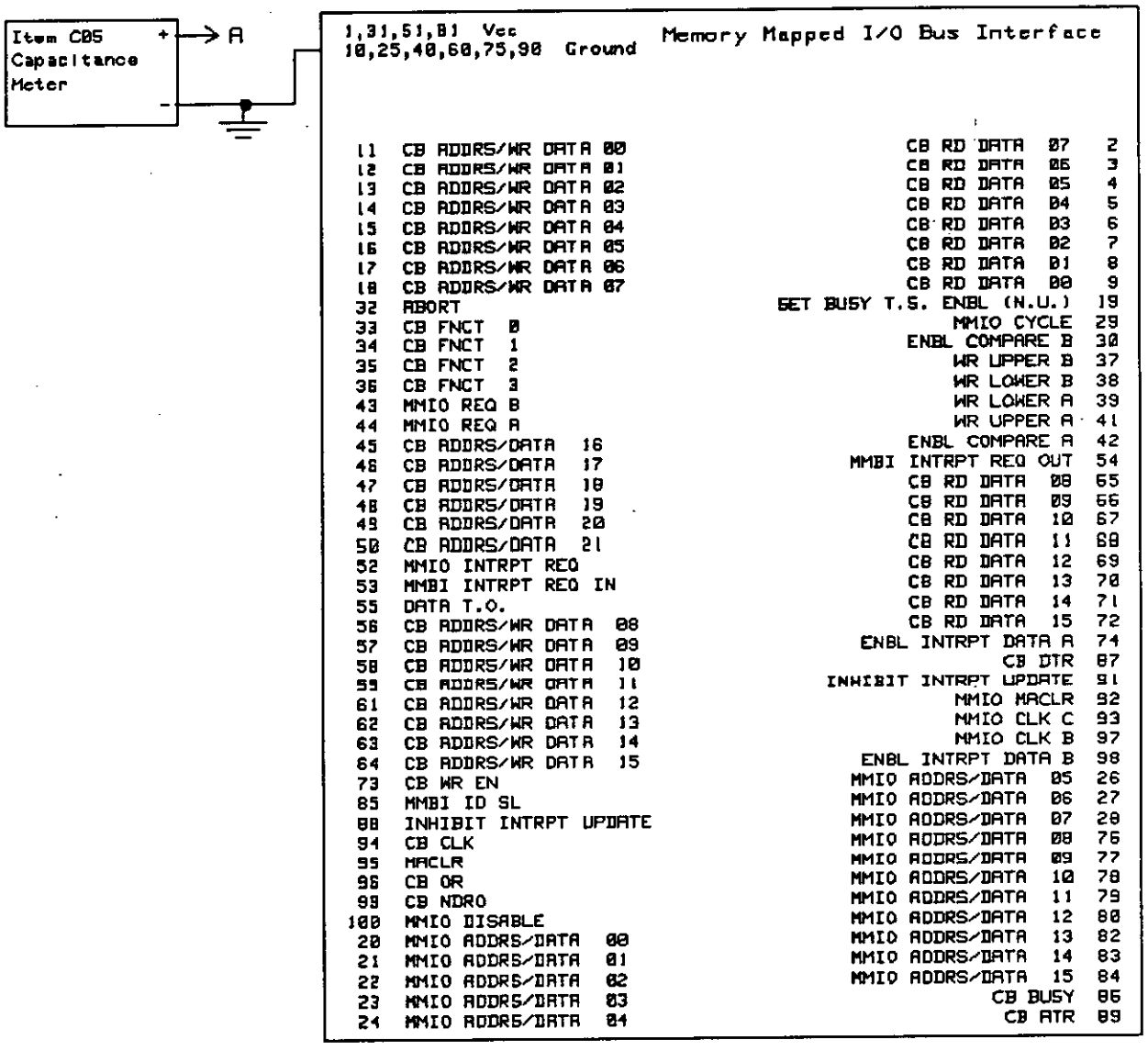


FIGURE 12. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1.8 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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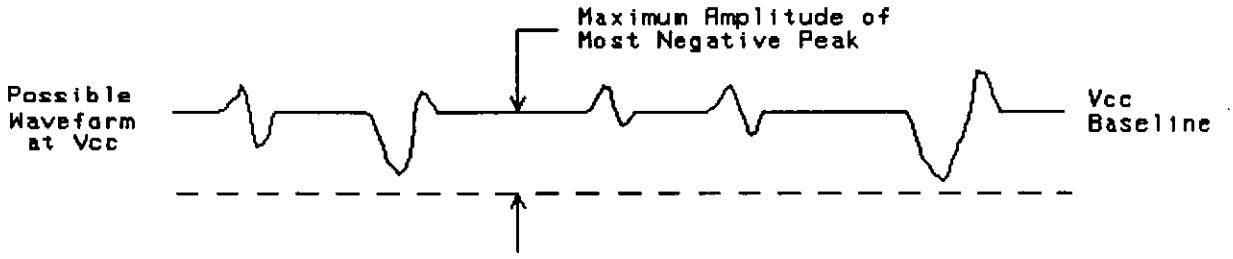
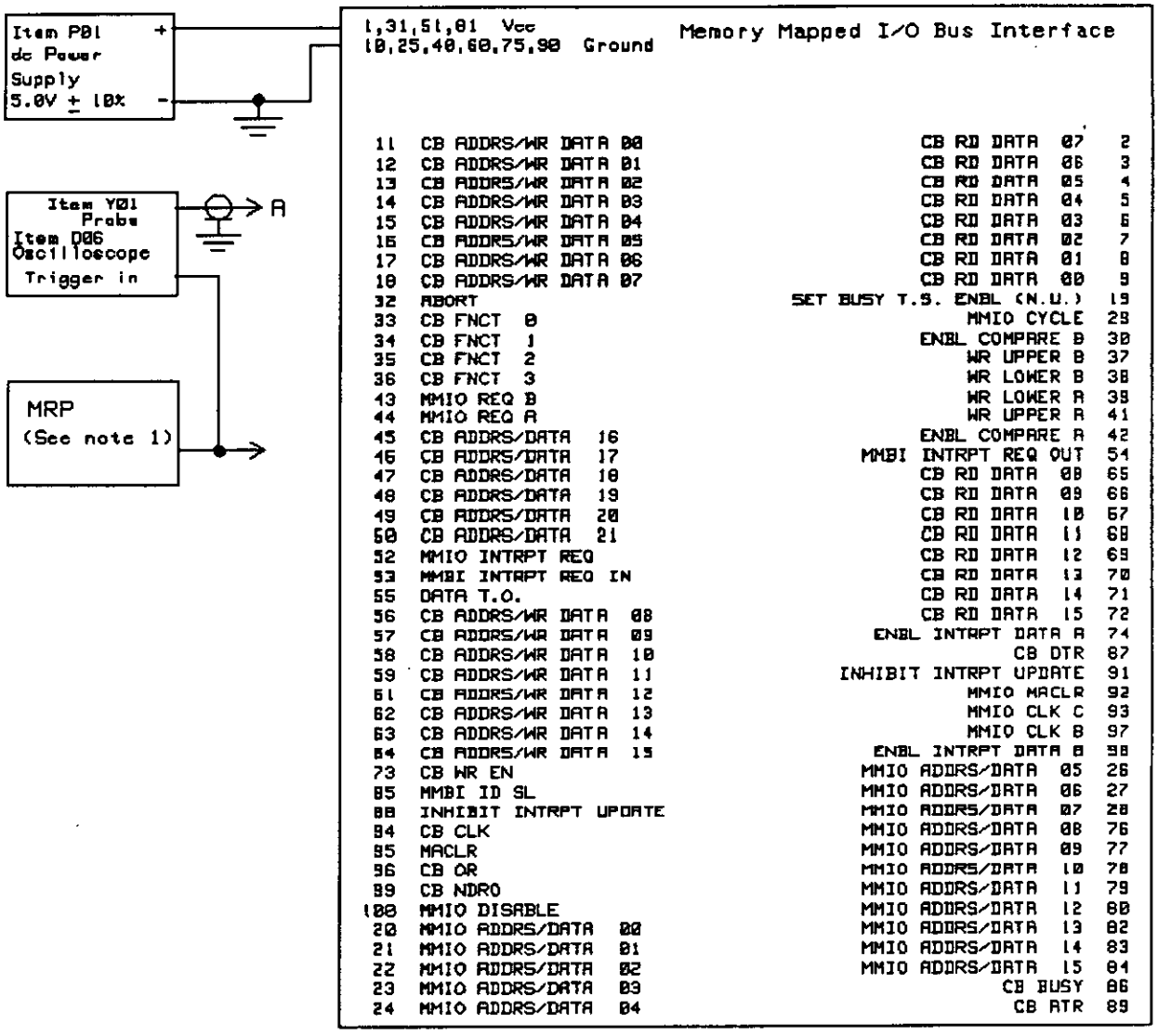


FIGURE 13. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XIX.

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